

RAG suppresses group 2 innate lymphoid cells

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Abstract

Antigen specificity is the central trait distinguishing adaptive from innate immune function. Assembly of antigen-specific T cell and B cell receptors occurs through V(D)J recombination mediated by the Recombinase Activating Gene endonucleases RAG1 and RAG2 (collectively called RAG). In the absence of RAG, mature T and B cells do not develop and thus RAG is critically associated with adaptive immune function. In addition to adaptive T helper 2 (Th2) cells, group 2 innate lymphoid cells (ILC2s) contribute to type 2 immune responses by producing cytokines like Interleukin-5 (IL-5) and IL-13. Although it has been reported that RAG expression modulates the function of innate natural killer (NK) cells, whether other innate immune cells such as ILC2s are affected by RAG remains unclear. We find that in RAG-deficient mice, ILC2 populations expand and produce increased IL-5 and IL-13 at steady state and contribute to increased inflammation in atopic dermatitis (AD)-like disease. Further, we show that RAG modulates ILC2 function in a cell-intrinsic manner independent of the absence or presence of adaptive T and B lymphocytes. Lastly, employing multiomic single cell analyses of RAG1 lineage-traced cells, we identify key transcriptional and epigenomic ILC2 functional programs that are suppressed by a history of RAG expression. Collectively, our data reveal a novel role for RAG in modulating innate type 2 immunity through suppression of ILC2s.

eLife assessment

This study provides new insights into the expression profile of ILCs that demonstrate a history of RAG expression. It examines in part the potential intrinsic regulation of RAG expression and seeks to understand how the epigenetic state of ILCs is established, although a full understanding of intrinsic factors is **incomplete**. The work provides an **important** molecular dataset, and with further strengthening of the understanding of intrinsic regulation, this paper would be of interest more broadly to cell biologists seeking to understand immune cell development.

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Introduction

Atopic disorders such as atopic dermatitis (AD), asthma, and food allergy are associated with T helper type 2 (Th2) cell responses, elevated production of the type 2 cytokines interleukin(IL)-4, IL-5, and IL-13, and induction of immunoglobulin(Ig)E^{1–4}. Classically, this allergic inflammatory cascade is believed to originate with antigenic stimulation of T cell receptors on adaptive T cells, which in turn results in the production of IgE from B and plasma cells capable of binding the same antigen. Indeed, presence of antigen-specific IgE reactivity is a hallmark across atopic disorders.⁵ Thus, for decades, antigen-specific adaptive Th2 cell responses have been the primary focus of investigation in the pathogenesis of atopic diseases. However, recent studies indicate that innate immune cells are sufficient to not only drive allergic pathology, but also amplify adaptive Th2 cell responses^{6–9}. These studies suggest that innate immune mechanisms may play a larger role in driving atopic inflammation than previously recognized.

Innate lymphoid cells (ILCs), while lacking antigen receptors generated by recombination activating gene proteins RAG1 and RAG2 (collectively called RAG), are the innate counterparts of T cells. For example, ILC2s mirror adaptive Th2 cells in their developmental requirements, cytokine profiles, and effector functions.¹⁰ Unlike classical T cells, ILC2s are concentrated at barrier surfaces to rapidly respond to microbial and environmental stimuli. ILC2s are key mediators of inflammatory skin conditions like AD^{11–13}. Indeed, in murine models of AD-like disease, type 2 skin inflammation can still occur despite the absence of adaptive T cells, but is further reduced after depletion of ILC2s^{12,13}. Furthermore, recent studies have shown that ILC2s harbor non-redundant functions in the presence of the adaptive immune system in the setting of anti-helminth immunity^{14,15}. These findings suggest that ILC2 dysfunction may also uniquely contribute to the pathogenesis of atopic diseases, independent of adaptive immunity. However, the cell-intrinsic mechanisms that drive ILC2 dysregulation remain poorly understood.

ILC2s were originally discovered due to their capacity to orchestrate multiple allergic pathologies in immunocompromised mice, most notably in RAG-deficient mice that lack T and B cells^{16–22}. These discoveries fundamentally redefined our understanding of allergic diseases and placed a major focus on ILC2s as potential drivers of human allergic disease. However, despite ILC2s not requiring RAG expression for their development, fate mapping studies in mice have demonstrated that up to 60% of ILC2s have historically expressed RAG1 during development^{23,24}. Although previous work has described roles of RAG beyond antigen receptor recombination in developing T and B cells^{25–27} and NK cells²⁴, how this developmental expression of RAG impacts ILC2s remains unclear.

By directly comparing RAG-deficient and RAG-sufficient mice, we unexpectedly found enhanced AD-like disease in RAG-deficient mice, despite the lack of adaptive lymphocytes to contribute to AD-like inflammation. Using splenocyte replenishment and bone marrow chimeras, we show that RAG suppresses ILC2 activation and expansion in a cell-intrinsic manner. Employing a RAG1-lineage reporter mouse line, we performed simultaneous single-cell multiomic RNA and ATAC sequencing to show that RAG fate-mapped ILC2s display unique transcriptional and epigenomic alterations consistent with the suppression of effector cytokine production. Collectively, our studies reveal evolutionarily conserved regulatory functions of RAG within innate lymphocytes, extending beyond the generation of antigen receptors in adaptive lymphocytes.

Results

RAG deficiency leads to expansion and activation of ILC2s

AD-like disease can be elicited in the skin of mice with repeated application of the topical vitamin D analog calcipotriol (MC903)²⁸. Although it has been previously demonstrated that MC903 can induce AD-like disease in RAG-deficient mice that lack T and B cells, in part via ILC2 activation^{12,13}, the relative contributions of ILC2s and the adaptive lymphocyte compartment have not been rigorously evaluated. We hypothesized that the presence of Th2 cells, in addition to ILC2s, would lead to enhanced AD-like disease in an additive fashion. In testing this, we evaluated both RAG1-sufficient wild-type (WT) mice and RAG1-deficient *Rag1*^{-/-} mice in the setting of AD-like disease (**Fig. 1A**). Unexpectedly, we observed that *Rag1*^{-/-} mice developed increased ear skin thickness (**Fig. 1B**) and higher frequencies of ILC2s in the skin-draining lymph nodes (sdLNs) compared to control WT mice (**Fig. 1C**, **S1A,B**). Furthermore, a larger proportion of ILC2s from *Rag1*^{-/-} mice exhibited production of both IL-5 (**Fig. 1D**, **S1C**) and IL-13 (**Fig. 1E**, **S1C**). Our findings indicated that RAG1 deficiency results in paradoxically worse AD-like disease in association with enhanced ILC2 expansion and activation.

To determine whether this phenomenon was specific to AD-like pathological conditions, we next examined the sdLNs in *Rag1*^{-/-} and lymphocyte-sufficient *Rag*^{+/-} littermate control mice in the absence of disease (**Fig. 1F**). We found that the frequency of ILC2s was increased at steady state in *Rag1*^{-/-} sdLNs (**Fig. 1G**) and that a higher proportion of these ILC2s produced both IL-5 (**Fig. 1H**) and IL-13 (**Fig. 1I**) compared to WT controls. The RAG recombinase requires both RAG1 and RAG2 components to successfully rearrange a functional antigen receptor in adaptive lymphocytes²⁹. Thus, to test whether our findings are specific to RAG1, or related to function of the overall RAG complex, we similarly examined the steady-state profile of ILC2s in *Rag2*^{-/-} mice (**Fig. S2A**). Deficiency of RAG2 led to an expansion of ILC2s in the sdLNs (**Fig. S2B**) and increased proportions of ILC2s expressing IL-5 (**Fig. S2C**) and IL-13 (**Fig. S2D**) similar to *Rag1*^{-/-} mice. Collectively, these findings suggest that the RAG recombinase modulates ILC2 function at steady state and during type 2 inflammation. However, whether the hyperactive ILC2 phenotype is due to a cell-intrinsic process or simply due to the absence of T and B cells was unclear.

ILC2 suppression by RAG is cell intrinsic

Given the importance of the adaptive lymphocyte compartment in shaping the secondary lymphoid organ repertoire, we next wanted to examine whether the presence of adaptive lymphocytes could restore ILC2 homeostasis in RAG-deficient mice. To test this, we created splenocyte chimera mice by reconstituting both *Rag1*^{-/-} and control WT mice with splenocytes containing T and B cells from WT donor mice (**Fig. 2A**). We first assessed the overall level of immune reconstitution in the recipient mice and found fully restored proportions of CD4⁺ (**Fig. S3A**) and CD8⁺ (**Fig. S3B**) T cells in the spleens of recipient *Rag1*^{-/-} mice, although B cell numbers remained significantly lower than in WT mice (**Fig. S3C**). Upon induction of AD-like disease, we found that the *Rag1*^{-/-} mice still exhibited increased ear skin thickness (**Fig. 2B**), enhanced expansion of ILC2s (**Fig. 2C**), and increased proportions of ILC2s expressing IL-5 (**Fig. 2D**) and

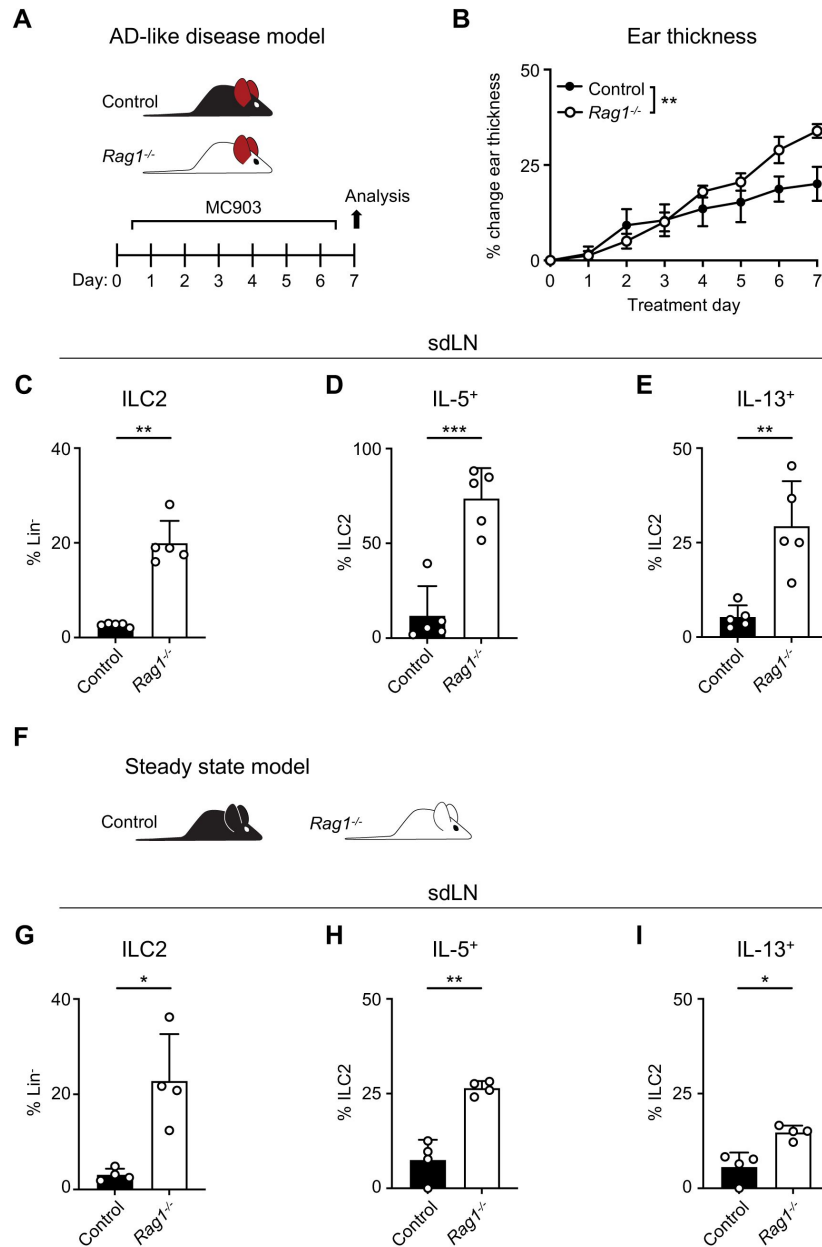


Figure 1

RAG deficiency leads to expansion and activation of ILC2s during inflammation and at steady state.

A) Experimental schematic of AD-like disease. WT (Control) B6 mice or *Rag1*^{-/-} mice treated topically to the inner surface of each ear with 2 nmol MC903 in 10 μ L ethanol vehicle daily for 7 days develop AD-like inflammation. **B**) Ear thickness measured daily in AD-like inflammation. Data representative of at least 2 independent experiments, 5 mice/group. ** $P < 0.01$ by 2-way ANOVA with Sidak's multiple comparisons test, day 7. **C**) Proportion of CD90⁺, Lin⁻ cells (Lin⁻ defined as CD3⁻, CD5⁻, CD11b⁻, CD11c⁻, CD19⁻, NK1.1⁻, and Fc ϵ R1⁻) determined to be ILC2s (IL-33R⁺) in skin-draining lymph nodes (sdLN) from WT or *Rag1*^{-/-} mice with AD-like ear inflammation. Percent ILC2 from sdLN in AD-like disease following PMA/iono stimulation positive for **D**) IL-5 or **E**) IL-13 staining. **F**) Schematic of steady state analysis of sdLN from WT (Control) or *Rag1*^{-/-} mice. **G**) Proportion of steady state sdLN CD90⁺, Lin⁻ cells determined to be ILC2s as in **C**). Percent ILC2 from sdLN in steady state following PMA/iono stimulation positive for **H**) IL-5 or **I**) IL-13 staining. **C-E**; **G-I**) Data representative of at least 2 independent experiments, 4-5 mice/group * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ by two-tailed Welch's t test. All data represented as mean with standard deviation.

IL-13 (Fig. 2E) in the sdLNs. Interestingly, we found significantly higher proportions of eosinophils in the spleens of *Rag1*^{-/-} recipient mice (Fig. S3D), possibly reflecting the increased IL-5 production we observed from ILC2s. These findings indicate that the mere introduction of exogenous T and B cells is not sufficient to suppress ILC2 dysregulation in the setting of RAG deficiency.

To further test whether this phenotype is mediated by cell-intrinsic RAG expression, we next generated mixed bone marrow (BM) chimeras. We harvested BM from congenic CD90.1⁺ WT and CD90.2⁺ *Rag1*^{-/-} donor mice on a CD45.2⁺ background and reconstituted sub-lethally irradiated CD45.1⁺ congenic WT recipients with a 50:50 mixture of WT:*Rag1*^{-/-} BM (Fig. 2F). After confirming reconstitution of donor immune cells in the sdLN (Fig. S4A,B), we examined the frequency and activity of ILC2s in the sdLNs based on whether they originated from WT (CD90.1⁺) or *Rag1*^{-/-} (CD90.2⁺) donors (Fig. S4C). Strikingly, of the total donor ILC2s, the majority were derived from *Rag1*^{-/-} donors (Fig. 2G). This was not due to differences in overall donor reconstitution, since measuring all Lin⁻ cells revealed WT donor cells outnumbered those from *Rag1*^{-/-} donors (Fig. S4D). Of the total IL-5- (Fig. 2H) and IL-13-expressing (Fig. 2I) ILCs, the majority were also derived from *Rag1*^{-/-} donors. Taken together, these data suggest that cell-intrinsic RAG activity in ILC2s may limit their capacity to expand and become activated.

A history of RAG expression marks a subpopulation of ILC2s in the skin draining lymph node

In contrast to resting naïve T cells, ILC2s resemble activated Th2 cells at steady state based on their transcriptomic and epigenomic profiles^{30,31}. While both T cells and ILC2s exhibit historical RAG expression²³, they do not actively express the protein in their mature state³². Taken together, these findings provoke the hypothesis that ILC2s are regulated by RAG early in development to imprint alterations that influence their activity as mature cells. To distinguish ILC2s as either having a history of RAG expression or not, we utilized a RAG lineage tracing system, whereby a *Rag1*^{Cre} mouse was crossed to a reporter mouse expressing tandem dimer red fluorescent protein (tdRFP) in a Cre-dependent manner from the *Rosa26* locus (Fig. 3A)^{24,33}. This system allowed us to compare RAG-experienced (RAG^{exp}) and RAG-naïve (RAG^{naïve}) lymphoid cells, including ILC2s, simultaneously originating from the same immunocompetent host, thus removing confounders inherent in knockout and chimera experiments. Analysis of sdLN from the reporter mice revealed that nearly all CD4⁺ T cells, CD8⁺ T cells, and B220⁺ B cells expressed tdRFP (positive history of Cre expression from the *Rag1* locus), consistent with the known requirement of RAG expression for their development (Fig. 3B-D, G). We also examined NK cells, since certain subsets of NK cells are known to express RAG during their development²⁴, and we observed that 60% of NK cells were tdRFP⁺, similar to previous findings (Fig. 3E, G)²⁴. In the ILC2 population of the sdLN, around 50% were tdRFP⁺ (Fig. 3F, G), similar to proportions of RAG fate-mapped ILC2s previously observed in the fat²⁴ and lung^{23,24}. These findings demonstrate that there are heterogeneous populations of ILC2s marked by differential tdRFP⁺ fate mapping. Importantly, this provided us with the possibility to profile these different ILC2 subsets based on *Rag1*^{Cre}-activated expression of tdRFP.

Multomic profiling enhances the detection of rare tissue-specific ILC2s

Transient RAG expression early in lymphoid development leads to well-characterized, durable effects on B and T cell development and function mainly through successful genomic rearrangement of antigen receptors. Yet, our data indicate that RAG expression also imprints phenotypic changes on ILC2s, which can develop independently of functional antigen receptors, provoking the hypothesis that RAG expression may affect broader epigenomic and transcriptional

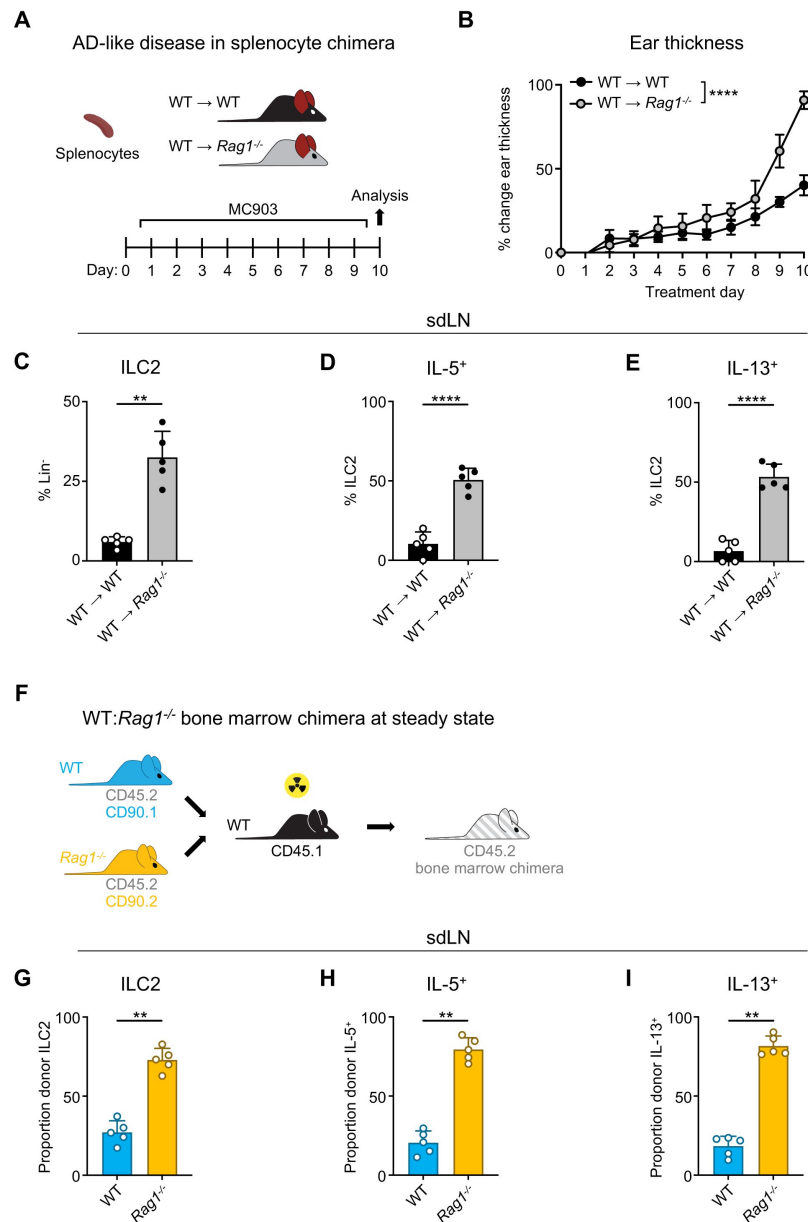


Figure 2

Homeostatic expansion and activation of RAG-deficient ILC2s is cell intrinsic

A) Experimental schematic of AD-like disease in splenocyte chimera experiment. WT B6 or *Rag1*^{-/-} mice received WT splenocytes and developed AD-like inflammation after subsequent topical treatment with 2 nmol MC903 in 10 μ L ethanol vehicle to each ear daily for 10 days. **B)** Ear thickness measured daily in AD-like inflammation. Data representative of 2 independent experiments, 4-5 mice per group. **** $P < 0.0001$ by 2-way ANOVA with Sidak's multiple comparisons test, day 10. **C)** Proportion of CD90⁺, Lin⁻ cells (Lin⁻ defined as CD3⁺, CD5⁺, CD11b⁺, CD11c⁺, CD19⁺, NK1.1⁺, and Fc ϵ R1⁺) determined to be ILC2s (IL-33R⁺). Percent ILC2 from sdLN in splenocyte chimera mice with AD-like disease after PMA/ionomycin stimulation positive for **D)** IL-5 or **E)** IL-13 staining. **F)** Schematic of bone marrow chimera experiment. Equal quantities of bone marrow cells from *Rag1*^{-/-} (CD45.2, CD90.2 – orange) and WT (CD45.2, CD90.1 – blue) C57Bl/6J donor mice were used to reconstitute the immune systems of irradiated recipient WT (CD45.1 – black) C57Bl/6J mice. **G)** Proportion of donor (CD45.2⁺) ILC2 defined as in (C) in sdLN by donor source (CD90.1⁺ – WT, CD90.2⁺ – *Rag1*^{-/-}). Proportion of Lin⁻ ILCs by donor source positive for **H)** IL-5 and **I)** IL-13 following PMA/ionomycin stimulation and cytokine staining. **C-E)** Data representative of at least 2 independent experiments, 4-5 mice per group. ** $P < 0.01$, **** $P < 0.0001$ by two-tailed Welch's t test. **G-H)** Data representative of at least 2 independent experiments, 4-5 mice per group. ** $P < 0.01$ by two-tailed ratio means paired t test. All data represented as mean with standard deviation.

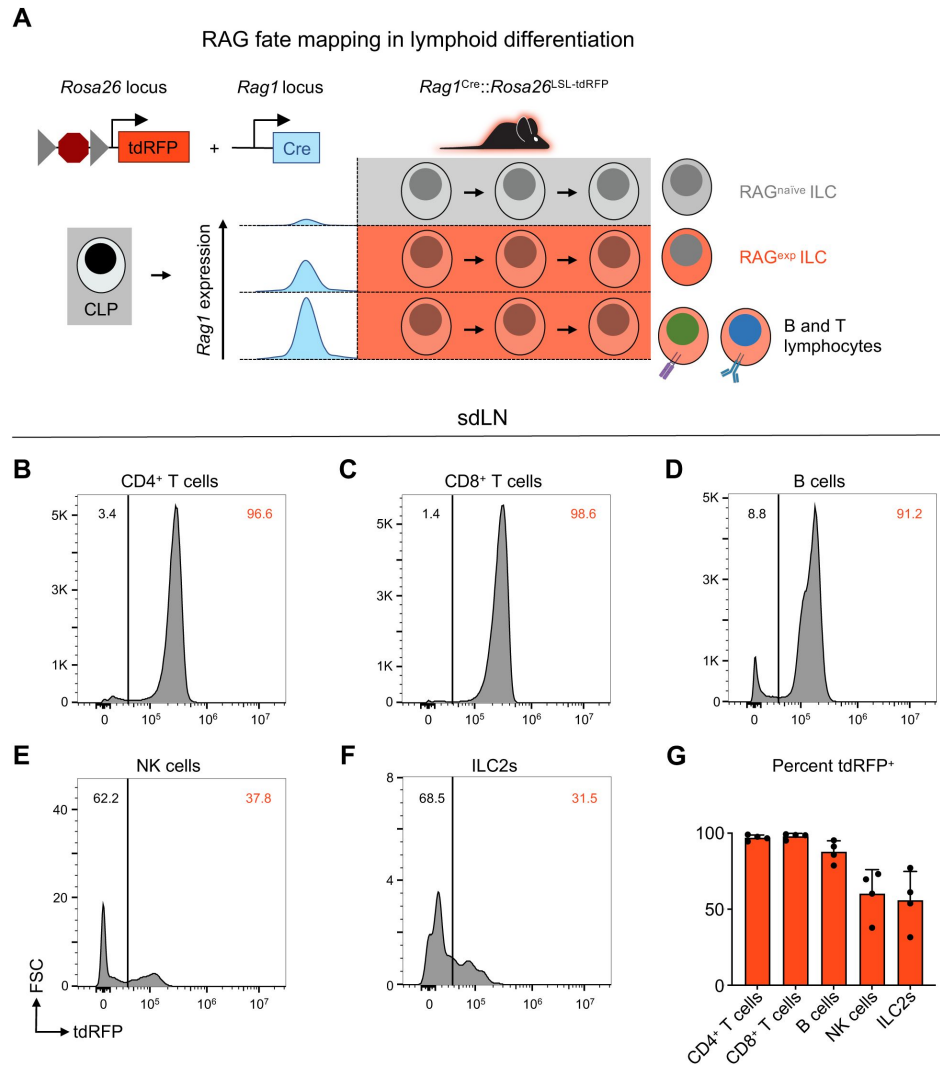


Figure 3

A history of RAG expression marks a population of ILC2s in the sdLN.

A) Schematic of RAG fate mapping in the lymphoid cell compartment using reporter mice expressing Cre-inducible tandem dimer red fluorescent protein (tdRFP) from the *Rosa26* locus crossed to mice expressing Cre recombinase from the *Rag1* locus. **B-F)** Histograms of tdRFP signal in CD45⁺ sdLN cells by cell type for **B)** CD4⁺ T cells (B220⁻, CD3⁺, CD4⁺), **C)** CD8⁺ T cells (B220⁻, CD3⁺, CD8⁺), **D)** B cells (MHCI⁺, B220⁺), **E)** NK cells (B220⁻, CD3⁻, CD4⁻, CD8⁻, CD49b⁺, NK1.1⁺), **F)** ILC2s (B220⁻, CD3⁻, CD4⁻, CD8⁻, CD49b⁻, NK1.1⁻, CD11b⁻, CD11c⁻, SiglecF⁻, CD90⁺, KLRG1⁺ or ICOS⁺ or IL-33R⁺), **F)** quantification of tdRFP⁺ proportion of each cell type.

programs. Furthermore, our data indicate that the impact of RAG on ILC2 function has implications for AD-like skin inflammation, suggesting a persistent effect that modulates phenotypes of type 2 inflammation.

To test these hypotheses, we performed combined single-nuclei RNA sequencing (snRNA-seq) and ATAC sequencing (snATAC-seq) of sDLN cells from RAG fate-mapped mice at steady state and in the setting of AD-like disease (**Fig. S5A,B**). Because fluorescence-activated cell sorting (FACS) can cause physical stress, cell loss, and contamination, which can introduce unwanted perturbations in target cells, instead we utilized gentle initial negative selection by magnetic activated cell sorting (MACS) to remove most B and T cells and monocytes prior to sequencing (**Fig. S5A**). This allowed us to enrich for innate immune cell populations prior to sequencing. Further, we used the gene encoding tdRFP as a barcode to differentiate between RAG^{exp} (RAG fate map-positive) and RAG^{naive} (RAG fate map-negative) ILC2s at the single-cell level (**Fig. 5A**). The multiomic data was analyzed using recently developed pipelines in Cell Ranger, Seurat^{34–36}, and Signac³⁷, and sequenced cells were further filtered computationally to enrich for ILCs, as in previous studies (see methods)³⁸.

In addition to gene expression (GEX) information derived from snRNA-seq (**Fig. 4A**), we calculated a “gene activity” (GA) score based on chromatin accessibility at gene loci (**Fig. 4B**) from the corresponding snATAC-seq dataset³⁷. Clustering the cells with each data subset alone and in combination using weighted nearest neighbor (WNN) analysis, we identified six clusters (**Fig. 4C**) that demonstrated consistent differences in cellular markers based on both metrics of GEX and GA (**Fig. 4D,E**; **Table S1**). Additionally, top markers for each cluster clearly differentiated each cell type (**Fig. 4F**). Despite successful ILC2 enrichment via MACS depletion for lineage markers and computational filtering (see methods), our data set included non-ILC2 populations determined by gene expression to be T cells, dendritic cells, B cells, and NK cells (**Fig. 4A,F**; **Table S1**), which allowed for broader multidimensional comparisons while studying this ILC2-enriched data set.

To further complement the GEX and GA assays, we utilized another method of detecting cell-specific marker genes, whereby chromatin regions that are differentially accessible (DA), or open, in each cluster could be linked by their physical proximity to specific genes (**Fig. 4G**, **S6**; **Table S2**; see methods). Comparing the top 100 GEX, GA, and DA markers in the ILC2 population, we identified a multiomic ILC2 signature of 235 unique genes (**Fig. 4H**; **Table S3,4**). While there was some overlap between each respective set, the multiomic approach enabled more extensive identification of an ILC2 gene program than through either snRNA-seq or snATAC-seq alone. The analysis revealed a variety of canonical ILC2-associated genes specific to the ILC2 cluster (**Fig. 4I**, **S6,7**) including the ILC2-activating cell surface receptors *Icos*^{39,40}, *Il2ra*^{41,42}, *Il18r1*^{38,43,44}, *Nmur*^{14,15,45–47}, and *Il1rl1* (encoding the receptor for IL-33)^{11,12,48,49}, as well as transcription factors such as *Gata3*^{19,50–53}, *Bcl11b*^{54–57}, *Maf*^{19,58,59}, *Ets1*⁶⁰, and *Rora*^{38,51,61–63}, all previously shown to be important for ILC2 development and/or function.

Expression of some secreted proteins can be difficult to capture in droplet-based snRNA-seq experiments due to low transcript levels and relatively shallow sequencing depth. With the addition of the complementary GA and DA assays from snATAC-seq, our analysis identified *Il5* (**Fig. 4I**, **S6B**), a canonical ILC2 cytokine, in the DA assay, while in the GA assay we found *Bmp7*^{64,65}, which has been shown to be secreted by ILC2s to influence browning of adipose tissue. Additionally, we identified the secreted chemokine *Ccl1* as an ILC2 marker gene^{66–69}, which along with its cognate receptor *Ccr8* (also an ILC2 marker in our analysis)^{44,67,70}, participates in a feed-forward circuit to drive ILC2 recruitment and expansion⁷¹. Thus, our findings demonstrate how genetic barcoding, combining transcriptomic and epigenomic analyses,

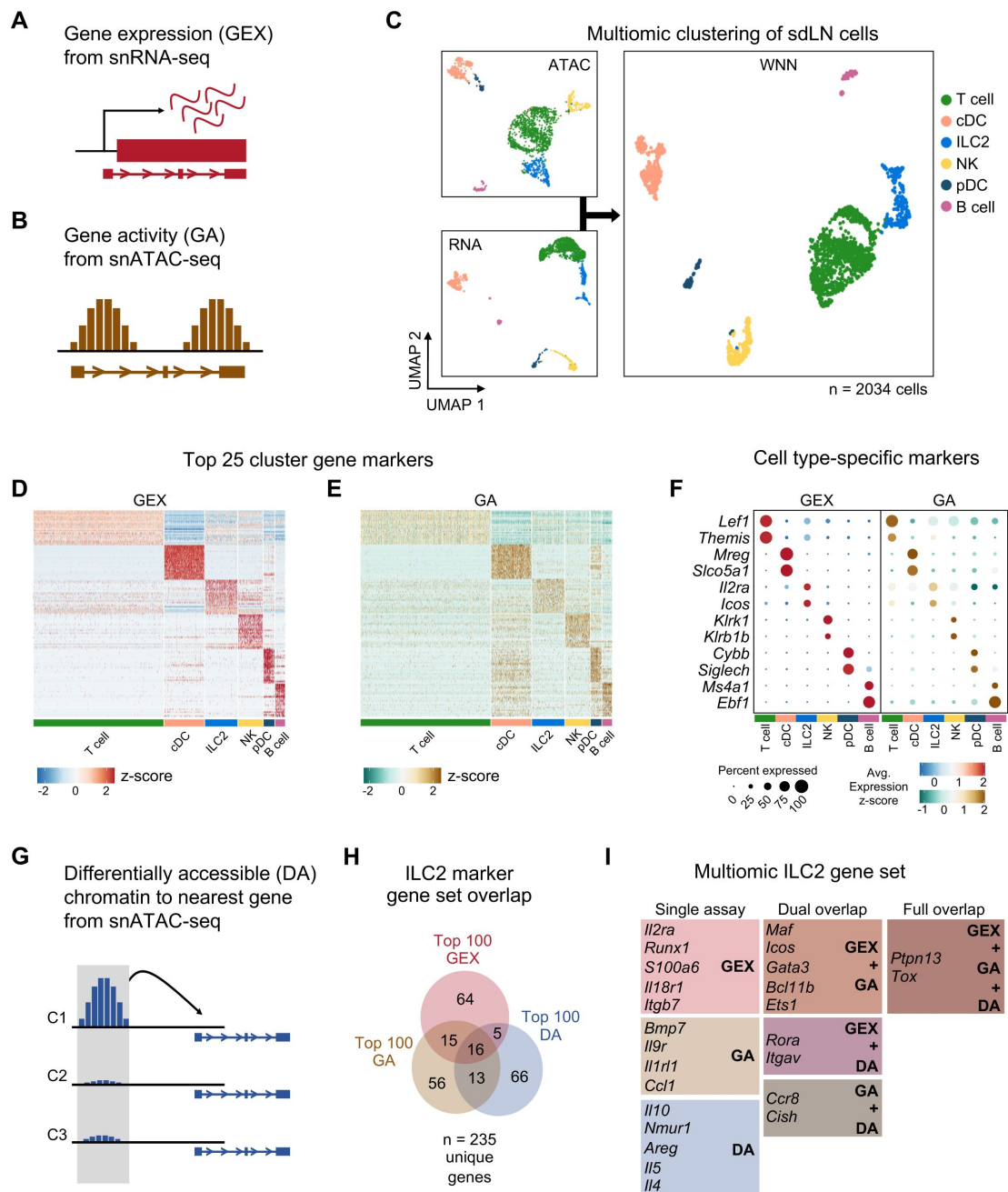


Figure 4

Multiomic analysis of ILC2s through single nuclei sequencing of the sdLN.

A) Schematic of the gene expression (GEX) assay derived from snRNA-seq data. **B**) Schematic of the gene activity (GA) assay, representing estimated transcription scores derived from snATAC-seq data using Signac. **C**) UMAP visualizations of independent analyses of RNA-seq and ATAC-seq data for 2034 sdLN cells after dimensional reduction and clustering combined using weighted nearest neighbor (WNN) analysis in Seurat. Cluster identities are color coded consistently throughout the following panels. Heatmaps of **D**) top 25 GEX marker genes and **E**) top 25 GA marker genes identified for each cluster. See **Table S1** for full lists of genes. **F**) Dotplots comparing selected marker genes for each cluster between the GEX and GA assays, with emphasis on known cell type-specific markers. **G**) Schematic of differentially accessible (DA) chromatin to nearest gene from snATAC-seq, which finds the nearest gene to any peak calculated to be differentially open in a particular cell cluster. See **Table S2** for full lists of top 25 DA cluster markers. **H**) Overlap of top 100 markers for the ILC2 cluster from the GEX, GA, and DA assays. See **Table S3** for top 100 DA peaks and distances to nearest genes and **Table S4** for full list of top 100 ILC2 markers. **I**) Selected genes from the ILC2 gene set for each assay individually and for overlaps.

and cross-validation across many published studies can yield new insights while providing internal control measures to elevate the rigor, robustness, and confidence of identifying gene signatures of rare populations such as ILC2s at the single-cell level.

ILC2s with a history of RAG expression are epigenomically suppressed

As noted above, barcoding the ILC2s afforded the opportunity to transcriptionally and epigenomically profile ILC2s under identical developmental conditions by dividing the ILC2 cluster into RAG^{exp} and RAG^{naïve} populations. We hypothesized that RAG^{exp} ILC2s would have a distinct transcriptional profile compared to ILC2s without any history of RAG expression. To test this, we calculated differentially expressed genes (DEGs) for the ILC2 cluster by *Rag1* fate-map status. Genes with higher expression in RAG^{exp} cells relative to RAG^{naïve} cells had positive fold change values, and vice versa, with genes relatively increased in RAG^{naïve} cells having negative values (Fig. S8A, Table S5). Using gene set enrichment analysis (GSEA)^{72,73} on the ranked list of DEGs, we found that gene sets generally representing lymphocyte activation and differentiation were suppressed in RAG^{exp} ILC2s compared to RAG^{naïve} ILC2s (Fig. S8B,C, Table S6), consistent with our previous observations that ILC2s are expanded and more activated in RAG-deficient mice relative to WT mice.

We next employed newly described methodologies^{37,74} that quantify associations between open chromatin peaks and the expression of nearby genes to describe the functional regulomes of both RAG^{exp} and RAG^{naïve} ILC2s (Fig. 5B). In this analysis, each ATAC peak can be linked to multiple genes, and each gene to multiple peaks, generating a list of “gene-to-peak links” or GPLs (see methods). For each gene, we interpreted the number of corresponding GPLs as a quantitative representation of the regulome activity for that gene. Considering RAG^{exp} and RAG^{naïve} cells as two separate populations, we generated two lists of GPLs (Table S7) defining functional regulomes for each population. We focused our analysis on the functional regulomes of ILC2s by filtering the GPL lists for the 235 unique ILC2 genes identified in our multiomic gene set (Fig. 4H; Table S8), then calculated the difference in GPLs between RAG^{exp} and RAG^{naïve} cells for each gene and sorted them; genes displaying greater numbers of GPLs in the RAG^{naïve} population are at the top, and genes with more GPLs in the RAG^{exp} population are at the bottom (Fig. 5C; Table S9).

We found that the ILC2 marker genes segregating toward the top of this list, corresponding to enhanced epigenomic activity in the RAG^{naïve} cells, tended to be genes previously identified to play positive roles in development, expansion, and activation of lymphoid cells. These included transcriptional regulators such as *Tox*^{75,78}, *Rora*^{38,51,61–63}, *Maf*^{49,58,59}, and *Gata3*^{19,50–52}, which are involved in early differentiation of both ILCs and lymphocytes (Fig. S9A). Indeed, epigenetic activation of the *Gata3* locus is recognized to play a critical role in development of both ILC2s⁷⁹ and Th2 cells^{80,81}. Additionally, surface receptors known to drive ILC2 activation upon stimulation including *Il18r1*^{38,43,44}, *Il1rl1*^{21,38,43,82,83}, and *Icos*^{39,40} had increased functional regulome activity in RAG^{naïve} ILC2s. In contrast, genes with more GPLs in the RAG^{exp} ILC2s tended to be associated with suppressive functions. For example, *Ndfip1* (Fig. S9B) encodes a regulatory protein that enhances activity of the ubiquitin ligase ITCH to negatively regulate inflammation^{84,85} and has been associated with asthma risk in GWAS studies⁸⁶. *Dusp1* partially mediates glucocorticoid effects through its ability to negatively regulate inflammation^{87,88}, is associated with eczema by GWAS⁸⁹, and has recently been shown to mark an anti-inflammatory set of ILCs⁶⁹. Last, *Asx1l* encodes a tumor suppressor that inhibits clonal hematopoiesis through its epigenomic regulatory effects in both mice and humans^{90–93}. Collectively, our GPL analysis stratifies the ILC2 gene signature based on RAG experience, where genes associated with ILC2 expansion and activation are poised in RAG^{naïve} cells, while genes associated with suppressive effects are poised in RAG^{exp} cells.

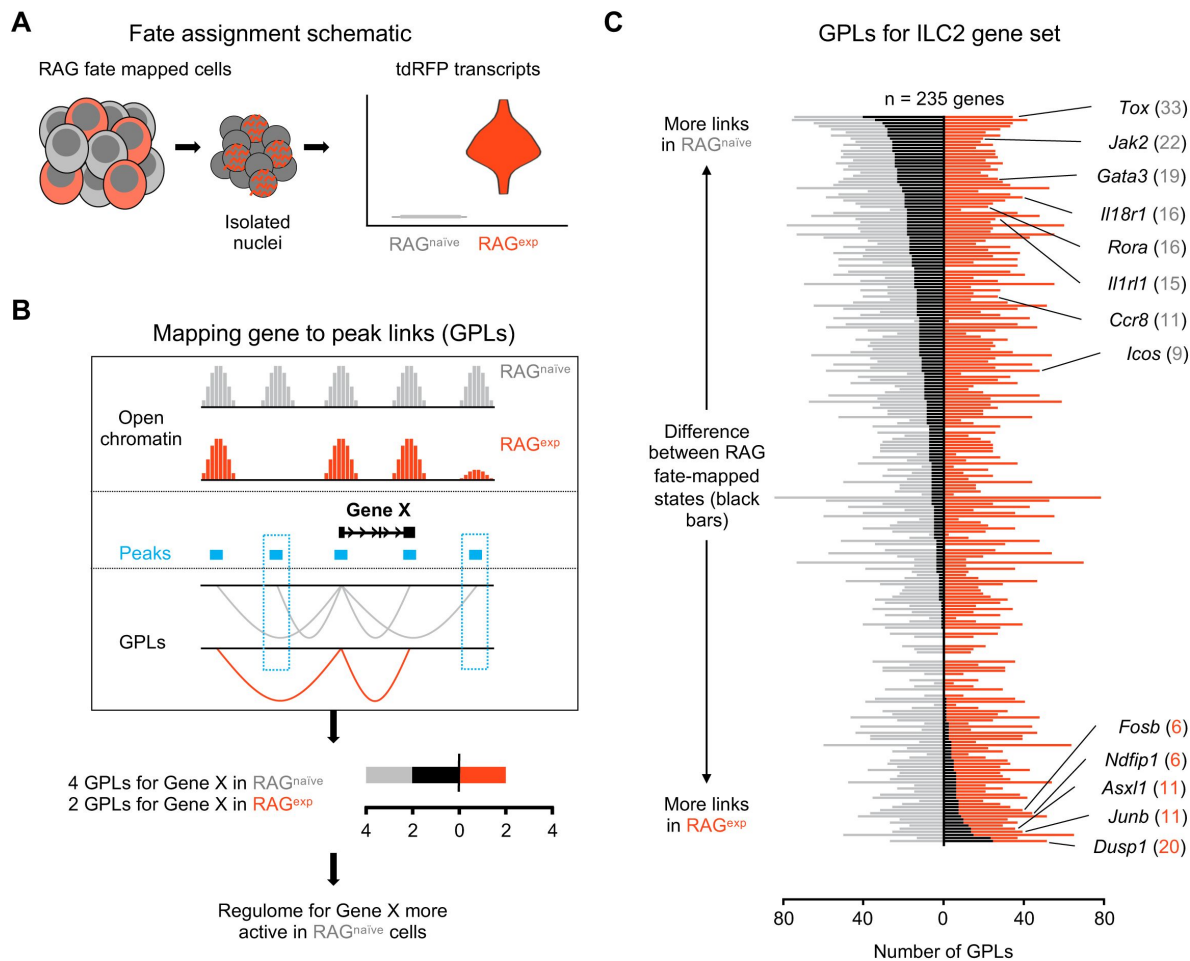


Figure 5

A history of RAG expression imprints transcriptomic and epigenomic modulation of ILC2 gene programs.

A) Schematic of transcriptional RAG fate mapping. Sequenced cells from the RAG fate map mouse (see [Figure 3A](#)) transcribe tdRFP only after Cre is expressed from the *Rag1* locus. Cells were assigned as either having a history of RAG expression (RAG^{exp} – tomato red) or not (RAG^{naive} – dark gray) based on detection of tdRFP transcript in the RNA-seq data. **B)** Schematic of mapping gene to peak links (GPLs). The LinkPeaks function of Signac (see methods) calculates significant correlations between open chromatin at defined peaks (teal bars) and nearby gene expression. These links represent inferred epigenomic-transcriptomic regulation, or “regulomes” based on the correlated snRNA- and snATAC-sequencing data. After calculating GPLs separately for each population (gray for RAG^{naive} and tomato red for RAG^{exp}), GPLs found in only one group, but not the other, can be identified (teal boxes). The difference in GPLs based on RAG experience for any given gene (e.g. Gene X) can be visualized on a bar graph, with the number of GPLs for RAG^{naive} (gray – left) and RAG^{exp} (red – right) plotted, with the difference overlaid as a black bar. **C)** GPLs calculated as in (B) for the multiomic ILC2 gene set identified in [Figure 4H](#) and [Table S4](#). All identified GPLs are listed in [Table S7](#), while ILC2 GPLs are listed in [Table S8](#). Genes are sorted from more links identified in the RAG^{naive} population at top to more links identified in the RAG^{exp} population at bottom. Select genes are labeled. Full ranked list by difference in GPLs is available in [Table S9](#).

A history of RAG expression modulates ILC2 epigenomes at steady state and in AD-like inflammation

Although our GPL analysis revealed a suppressive effect of RAG expression on ILC2 gene programs, we did not account for the additional variable of disease state in the initial analysis. To test whether RAG expression promotes a suppressive epigenomic program in ILC2s that is durable in the setting of inflammation, we first recalculated GPLs after splitting our dataset by both history of RAG expression (naïve vs. experienced) and disease (steady state vs. AD-like disease) to yield four lists of GPLs (**Fig 6A** [↗](#); **Table S10**). When we examined the intersection, or overlap, of peaks from ILC2 GPLs (**Table S11**), several notable patterns emerged (**Fig. 6B** [↗](#)). First, the largest set of peaks was shared by all RAG^{naïve} cells (gray bar), regardless of disease state, with the next largest peak sets belonging to either steady state or AD-like disease in the RAG^{naïve} cells. Second, there was a large set of peaks shared by all RAG^{exp} cells (red bar). Third, the intersections corresponding to disease states (steady state – yellow, AD-like disease – dark red), had relatively few unique peaks. These findings suggest that early exposure to RAG expression plays a larger role in modulating the epigenomic signature of the ILC2 gene program than exposure to disease. To confirm that the patterns we observed represent a specific effect of RAG expression on the ILC2 gene program, we performed the same analysis on GPL peaks for all genes in the dataset (**Fig. S10**). In contrast to the ILC2 gene set, the majority of GPL peaks for all genes was shared among all cell populations, consistent with epigenomic regulation of most genes being minimally affected by either RAG expression or AD-like disease. Last, in the ILC2 gene set analysis, we noted a set of poised peaks shared by all RAG^{naïve} cells and RAG^{exp} cells in the setting of AD-like disease, but not with RAG^{exp} cells at steady state (blue bar, **Fig. 6B** [↗](#)). We reasoned that this condition might capture some genomic loci that are suppressed by a history of RAG expression at steady state but are induced during inflammation.

Thus, we next quantified and sorted these GPLs to generate a list of genes with the most peaks “induced” during AD-like disease (**Fig. 6C** [↗](#), **Table S12**). Among the identified genes, we selected *Rora* (**Fig. 6C** [↗](#)) and *Ccr6* (**Fig. 6D** [↗](#)) to examine more closely for evidence of epigenomic activation in AD-like disease, given the role of these genes in ILC2 expansion^{61 [↗](#), 62 [↗](#)} and homing to sites of inflammation^{44 [↗](#), 94 [↗](#)}, respectively. For both genes, we observed more widespread open chromatin over the genomic region in the RAG^{naïve} cells compared to the RAG^{exp} cells, but this difference was partially abolished by increased open chromatin in AD-like disease in the RAG^{exp} cells. Taken together, our analysis reveals that a history of RAG expression selectively modulates activity of ILC2 gene programs across both steady state and during AD-like inflammation, while some programs are more evident at steady state given the uniquely poised nature of ILC2s.

RAG suppresses the Th2 locus

Our functional data demonstrate a role for RAG expression in regulating ILC2 development and activation, including limiting proportions of IL5⁺ and IL-13⁺ ILC2s at steady state and in AD-like disease. Prior work identified epigenomic priming in ILC2s early in development at the Th2 locus (comprised of the *Il4*, *Il13*, *Rad50*, and *Il5* gene loci) to enable rapid transcriptional responses during inflammation^{30 [↗](#)}. Thus, we hypothesized that RAG promotes the functional observations in ILC2s by suppressing the establishment of an active regulome at the Th2 locus. To test this hypothesis, we analyzed the Th2 locus in our multiomic data in greater detail. Using a similar strategy to our analysis of ILC2 marker genes, we calculated the number of GPLs in the RAG^{exp} and RAG^{naïve} cells, respectively, for the genes in the Th2 locus. (**Fig. 7A** [↗](#), **Table S7**). We found many GPLs associated with the four Th2 locus genes, including significant crosstalk *between* these genes, similar to previous observations (**Fig. 7A** [↗](#))^{95 [↗](#)–97 [↗](#)}. Importantly, we identified fewer GPLs in the RAG^{exp} cells, particularly for the *Il5* and *Il13* loci (**Fig. 7A** [↗](#)). As in our analysis of the ILC2 marker GPLs, we quantified the differences based on RAG fate mapping and found that all

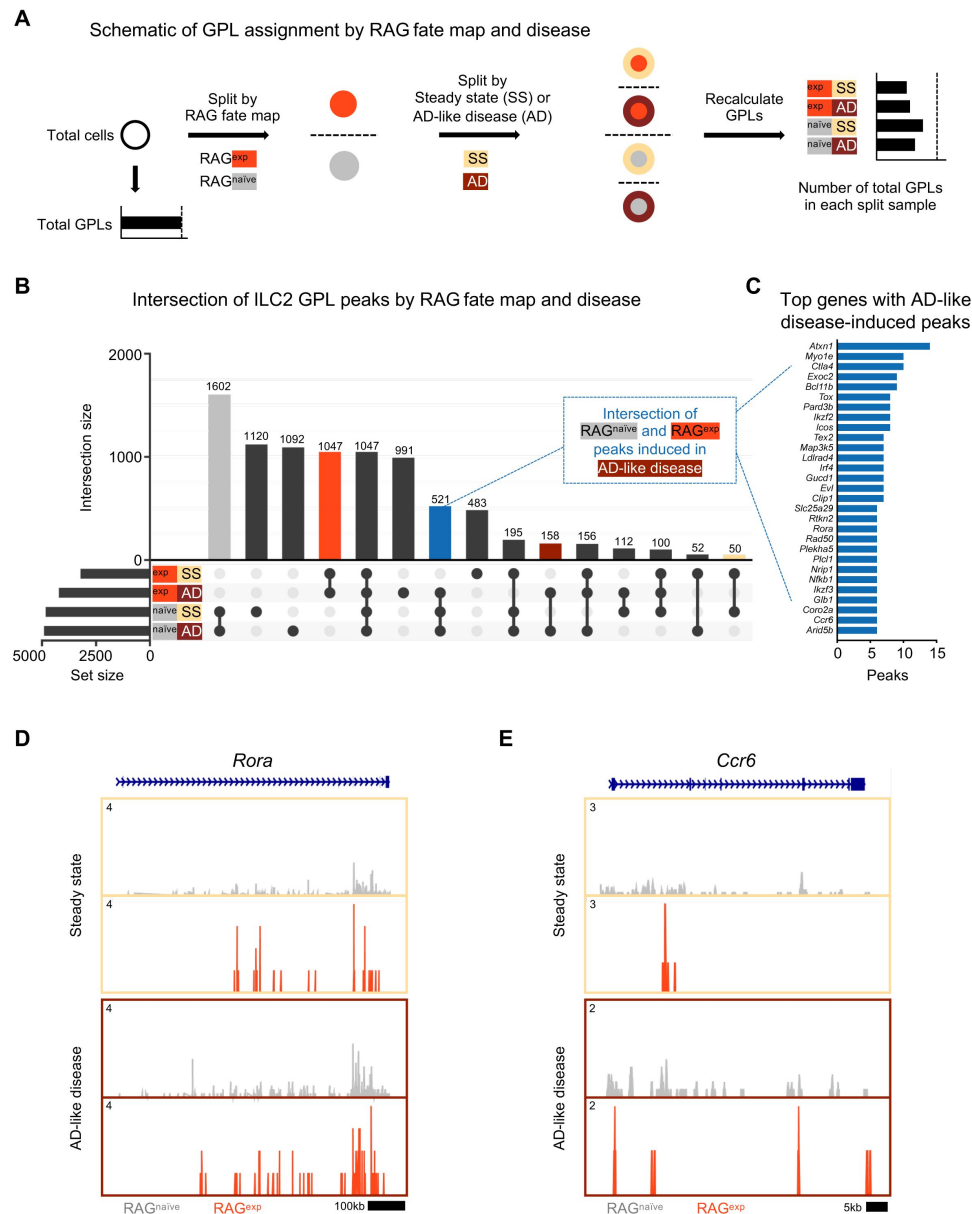


Figure 6

A history of RAG expression broadly influences ILC2 genes at steady state and in AD-like inflammation.

A) Schematic of the process to determine contribution of RAG fate map and disease states to GPLs for subsequent intersection analyses. GPLs were first calculated for all indicated cells, regardless of disease state or fate map. Cells were then split, first by RAG fate map (RAG^{exp} and RAG^{naive}), and again by disease state (SS – steady state, AD – AD-like inflammation). GPLs were recalculated for each split sample and matched back to the original set of total GPLs. **B)** UpSet plot visualizing intersections of peaks identified from ILC2 GPLs for split samples. Each row represents one of the four sets, and each column corresponds to an intersection of one or more sets (see methods). See **Table S10** for full list of GPLs for all genes. **Table S11** lists total and ILC2 peaks used for intersection analyses in each of the four sets. Columns identifying key intersections are color coded by the corresponding RAG fate map or treatment groups. The blue column indicates the intersection of peaks from RAG^{naive} cells and peaks induced by AD-like disease in RAG^{exp} cells. **C)** Top genes with the most AD-like disease-induced peaks. Peaks from the intersection between RAG^{naive} cells and inflamed RAG^{exp} cells were identified in corresponding GPLs, and genes were ranked by number of linked peaks identified. See **Table S12** for full list of ranked genes and associated GPLs. Open chromatin in the ILC2 cell cluster split by disease state and then by RAG fate map for the genomic loci of **C)** *Ccr6* and **D)** *Rora*.

genes in this locus had increased GPLs in RAG^{naïve} cells relative to RAG^{exp} cells (Fig. 6B; Table S13). Thus, our epigenomic data are consistent with a role for RAG expression in suppressing the type 2 regulome at the Th2 locus in ILC2s.

We next considered the additional effects of AD-like inflammation on the Th2 epigenomic regulome using the same approach we used to analyze the ILC2 gene set in Fig. 6. Again, we found the largest set of peaks was shared by the RAG^{naïve} cells, regardless of disease state, with the next largest peak sets belonging to either steady state or AD-like disease in the RAG^{naïve} cells (Fig. 7C). Furthermore, there was a large proportion of peaks shared by both RAG^{exp} cells, consistent with a major contribution of a history of RAG expression to epigenomic modulation of the Th2 regulome. To quantify the potential effect of AD-like inflammation on reversing RAG-mediated suppression of Th2 locus genes, we mapped the 14 peaks shared by RAG^{naïve} cells and RAG^{exp} cells in the setting of AD-like inflammation (i.e. only suppressed in RAG^{naïve} cells at steady state) back to the Th2 genes via their respective GPLs (Fig. 7C –blue bar, Table S14). Interestingly, *Il13*, which was not identified as a top ILC2 marker in our earlier analyses, had the highest number of linked peaks associated with potential induction in AD-like disease (Fig. 7D). When we examined the *Il13* locus in the ILC2 cluster more closely, we found more widespread open chromatin in the RAG^{naïve} cells compared to the RAG^{exp} cells (Fig. 7E). However, in the AD-like disease sample, the RAG^{exp} cells displayed increased open chromatin relative to the steady state, consistent with induction in the setting of inflammation, like our earlier findings for ILC2 genes such as *Ccr6* (Fig. 6E). Taken together, our functional data and multiomic analyses demonstrate a role for RAG expression in modulating genes critical for ILC2 development and function, including the key type 2 cytokines expressed from the Th2 locus.

Discussion

RAG recombinases evolved nearly 500 million years ago from endogenous transposons, crucially enabling antigen receptor rearrangement and emergence of the adaptive immune cell lineages present in all modern vertebrates^{29,98}. Indeed, RAG deficiency leads to a complete lack of B and T lymphocytes, manifesting clinically as severe combined immunodeficiency (SCID)^{99–101}. However, fate mapping studies have shown that multiple mature immune cell populations other than adaptive B and T lymphocytes have a history of RAG expression^{23,24,33,102,103}. More recent studies by Karo et al found that RAG expression during NK cell development influences multiple cellular functions including antitumor cytotoxicity, cell proliferation, and survival²⁴. Yet whether RAG modulates cellular functions of innate immune cell populations other than NK cells remains poorly understood. Here, using RAG-deficient mice, RAG fate mapping mice, and multiomic analyses, we report that RAG suppresses developmental and effector functions of ILC2s.

Our functional data in RAG-deficient mice demonstrate that populations of ILC2s producing the type 2 cytokines IL-5 and IL-13 preferentially expand in the absence of a history of RAG expression. This implies a specific role for RAG in developmental repression of ILC2s. Building on this, our multiomic RAG fate mapping analyses of ILC2 gene programs demonstrate extensive epigenomic differences between RAG^{exp} and RAG^{naïve} cells. We found RAG-associated epigenomic suppression at multiple functional levels, including cell surface receptors, key transcription factors, and the Th2 locus encoding the type 2 cytokines *Il5* and *Il13*. Although RAG is only transiently expressed early in lymphoid development¹⁰⁴, our data demonstrate that RAG expression can imprint durable effects on ILC2 gene programs to restrain their function.

It is increasingly recognized that expression of effector molecules for both ILCs and their counterpart adaptive lymphocytes (e.g. IL-13 from ILC2s and Th2 cells) is governed by finely tuned transcriptomic and epigenomic regulomes^{30,105–107}. ILCs tend to adopt these regulomes earlier in their development than T cells, and these “poised” regulatory elements are thought to

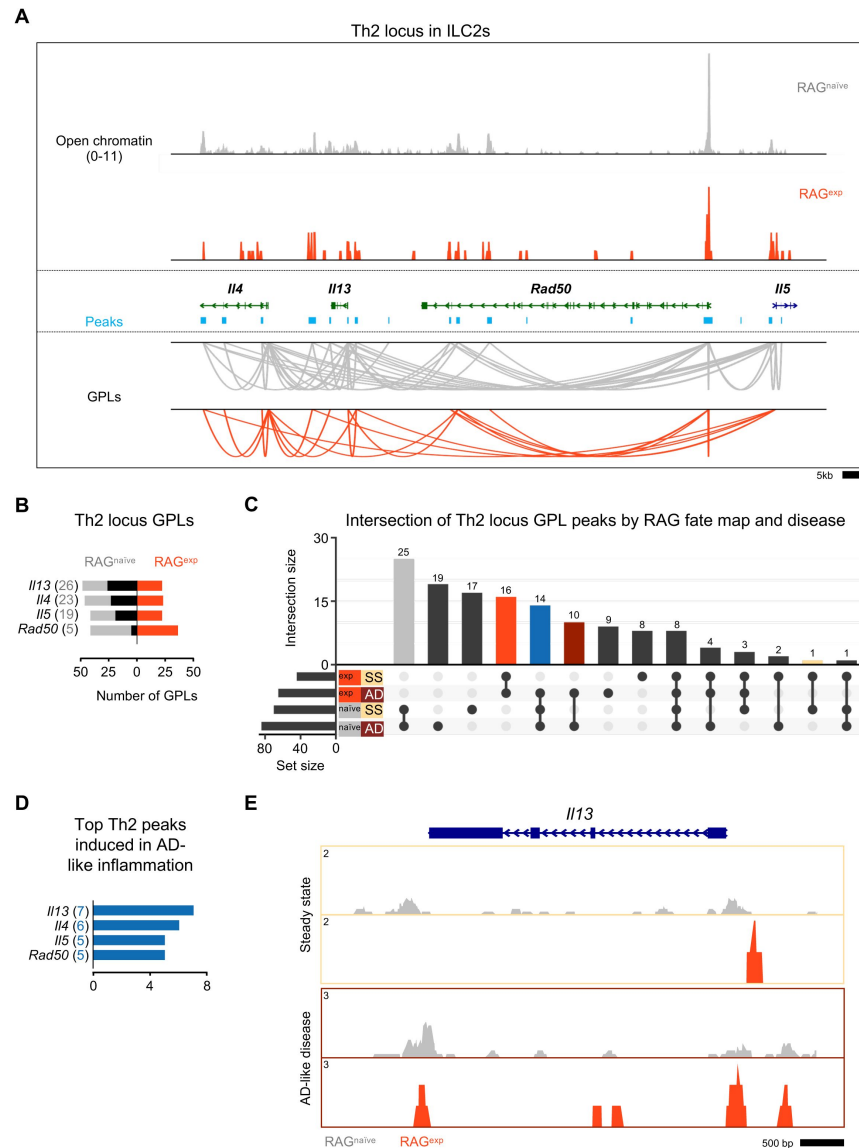


Figure 7

RAG suppresses the Th2 locus.

A) Coverage plot of the Th2 genomic locus. Open chromatin in the ILC2 cluster for each *Rag1* fate-mapped state is shown on top, and corresponding peaks (teal) and gene to peaks links (GPLs) are shown below for the RAG^{naive} sample (gray) and the RAG^{exp} sample (tomato red). Only GPLs that fit in the coverage window are shown. **B)** All GPLs identified in each fate map state for the Th2 locus genes *Il4*, *Il13*, *Rad50*, and *Il5*. See **Table S13** for full list of Th2 GPLs. The number of GPLs for each gene is shown on the left in gray for RAG^{naive} and on the right in tomato red for RAG^{exp} . The difference is superimposed in black, and genes are sorted from more GPLs identified in RAG^{naive} at top to more links identified in RAG^{exp} at bottom. **C)** UpSet plot of intersections of peaks identified from Th2 locus GPLs. GPLs were recalculated, this time for samples separately by both RAG fate map status (RAG^{exp} and RAG^{naive}) and disease (SS – steady state, AD – AD-like inflammation). Each row represents one of the four sets of peaks, and each column corresponds to an intersection of one or more sets (see methods). See **Table S11** for full list of peaks from GPLs for all genes, including Th2 genes, in each of the four sets. Columns identifying key intersections are color coded by the corresponding RAG fate map or disease groups. The blue column indicates the intersection of peaks from RAG^{naive} cells and peaks induced in AD-like disease in RAG^{exp} cells. **D)** Th2 genes sorted by number of AD-like disease-induced peaks. Peaks induced by AD-like disease were identified in corresponding GPLs, and genes were ranked by frequency of links to induced peaks (representation in identified GPLs). See **Table S14** for full list of ranked Th2 locus genes and associated GPLs. **E)** Open chromatin tracks, split by disease (beige box – steady state; maroon box – AD-like disease) and by RAG fate map (RAG^{naive} – gray, RAG^{exp} – red) for *Il13*.

underlie the ability of tissue-resident ILCs to rapidly respond to stimuli. In contrast, the regulomes of naïve T cells remain relatively inactive until stimulation. Given that T cells are uniformly RAG-experienced, our data provoke the hypothesis that RAG^{exp} ILC2s adopt a phenotype closer to that of naïve T cells and may require stronger stimuli than RAG^{naïve} ILC2s to become activated. Indeed, our analyses found the RAG-associated suppressive programs could be overcome in the setting of AD-like inflammation. Thus, sufficient RAG expression may mediate key events underlying the establishment and maintenance of functional regulomes not only in ILCs, but also T cells. How RAG might affect these changes, and whether they are independent of its enzymatic activity and/or antigen receptor recombination, remains to be elucidated.

Clinically, a link between enhanced type 2 immune activity and RAG dysfunction is well-established. Omenn Syndrome (OS) is a form of SCID characterized by exaggerated type 2 immune activation and typically arises in the setting of hypomorphic RAG gene mutations. Impaired antigen receptor rearrangement, with rare “leaky” recombination events, leads to expansion of autoreactive oligoclonal T cells, eosinophilia, and markedly elevated IgE^{108–112}. Similar phenotypes have been observed in mice harboring RAG mutations analogous to those found in human patients with OS^{113,114}. Notwithstanding these findings, the mechanisms underlying the propensity of oligoclonal T cells with hypomorphic RAG activity to preferentially develop into the Th2 subtype are unclear. Prior studies have found a role for regulatory T cells in controlling type 2 skewing of transferred T cells in RAG-deficient hosts, potentially explaining similar observations in patients with OS¹¹⁵. Our data provide an additional mechanism by which RAG dysfunction may lead to OS through loss of cell-intrinsic RAG-mediated suppression of type 2 cellular programs. Additionally, increased type 2 cytokine production from RAG-deficient ILC2s may, in *trans*, enhance expansion of the oligoclonal Th2 cell populations, IgE induction, and eosinophilia observed in RAG-deficient states like OS. However, whether other immune cell types with RAG dysfunction, such as ILCs, contribute to the pathogenesis of OS in humans has not been investigated.

Lymphoid acquisition of RAG activity may represent a newer evolutionary mechanism that fine tunes ancient innate immune cell programs in addition to enabling development of relatively newer antigen-specific adaptive immune cell populations. Independent of antigen receptor diversity, loss of this function may offer an explanation as to why oligoclonal T cells tend to expand and skew towards a Th2 cell phenotype in the setting of hypomorphic RAG function as in OS¹¹⁵. Further studies are needed to define whether the suppressive effects of RAG expression operate similarly in T and B cells. Although we demonstrate that this phenomenon is observed in ILC2s, whether hypomorphic RAG expression in bona fide Th2 cells not only results in oligoclonality but also loss of suppression of the Th2 locus independently of antigen receptor rearrangement remains an outstanding question. Indeed, during development of gene therapy strategies for RAG-deficient SCID, lower doses of wild type RAG transgene expression have been associated with development of OS-like conditions in transplanted RAG-deficient mice^{116–119}.

A major limitation of our study is lack of a defined mechanism for how RAG expression imprints durable epigenomic and transcriptomic changes in ILC2s. The mechanisms by which RAG mediates VDJ recombination are well-defined, from the biochemical details of DNA-binding to the epigenomic accessibility of antigen receptor loci and timing of RAG expression^{29,120–122}. Notwithstanding genomic stress²⁴ or potential RAG dose effects^{116–119}, how RAG expression might modulate broad developmental and functional lymphoid programs other than V(D)J recombination remains unclear. The RAG complex can bind both DNA and modified histones and has been observed to occupy thousands of sites across the genome²⁷. Thus, RAG may directly influence open chromatin states or obscure transcription factor binding sites to alter ILC2 development and function. Interestingly, RAG preferentially binds near transcription start sites of open chromatin in mouse pre-B cells, although corresponding effects on gene expression were not observed²⁷. Although canonical recombination sites are concentrated in the antigen receptor loci, cryptic recombination sites in other regions may be deleted or rearranged by RAG activity,

altering transcriptional regulation of associated genes²⁷. Finally, through its E3 ubiquitin ligase activity²⁹, RAG may influence immune signaling pathways independently of transcription altogether. Beyond basic mechanisms of how RAG modulates ILC2 development and function, whether these effects may contribute to pathologic processes modulated by ILC2s in organs other than the skin remains unclear.

Our studies expand prior work implicating RAG in critical immune functions beyond antigen receptor rearrangement that is exclusive to adaptive lymphocytes. Further, we provide additional insights into why patients with OS exhibit atopic syndromes in the setting of adaptive lymphocyte deficiency. Future studies into mechanisms underlying these findings may lead to new therapeutic avenues for disorders such as atopic dermatitis, food allergy, and asthma.

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Supervision, B.S.K.

Funding Acquisition, B.S.K.

Declaration of interests

B.S.K. is founder of Alys Pharmaceuticals; he has served as a consultant for 23andMe, ABRAX Japan, AbbVie, Amgen, Cara Therapeutics, Clexio Biosciences, Eli Lilly and Company, Escient Pharmaceuticals, Evommune, Galderma, Genentech, LEO Pharma, Pfizer, Recens Medical, Regeneron, Sanofi, Septerna, Triveni Bio, and WebMD; he has stock in ABRAX Japan, Alys Pharmaceuticals, Locus Biosciences, and Recens Medical; he holds a patent for the use of JAK1 inhibitors for chronic pruritus; and he has a patent pending for the use of JAK inhibitors for interstitial cystitis. A.M.V. has contributed to scientific advisory boards at Galderma and has performed sponsored research for Amgen.

Figure titles and legends

Materials and methods

Animal studies

Wild-type (WT) C57BL/6J and WT congenic strains (CD90.1, CD45.1), *Rag1*^{-/-}, and *Rag2*^{-/-} mice were initially purchased from the Jackson Laboratory and bred in house. The RAG fate-mapping strain *Rag1*^{Cre::Rosa26^{LSL}-tdRFP} was originally created in the lab of Paul Kincade³³ and bred in house. All mice were housed in specific-pathogen-free condition and environmentally controlled animal facility with a 12-hour light-dark cycle and given unrestricted access to food and water at Icahn School of Medicine at Mount Sinai or Washington University School of Medicine in St. Louis. All animal protocols and experiments were approved by the Institutional Animal Care and Use Committee (IACUC) at Icahn School of Medicine at Mount Sinai or Washington University School of Medicine in St. Louis. Experiments were performed on independent cohorts of male and female mice. For induction of AD-like disease, 8-to 12-week-old mice were treated with 2 nmol calcipotriol (MC903, Tocris Bioscience) in 10 μ L of 100% ethanol (EtOH) vehicle, or vehicle alone, on the bilateral ear skin daily for 7-10 days. Body weight and ear thickness were measured daily with a digital scale and analog caliper by the same investigator. For tissue harvest, animals were euthanized by CO₂ inhalation.

Flow cytometry

Cervical skin draining lymph nodes (sdLN) were removed from the mice and immediately homogenized manually through a 100 μ m cell strainer (Fisher Scientific) into a 50 mL tube with the end of a plunger from a 3 mL syringe. The strainer was washed with wash medium (2% vol/vol FBS/PBS) and the strained cells were centrifuged at 400g for 5 minutes at 4°C. Lymph node cell samples were stained with Zombie NIR viability dye (Biolegend; 1:500) to exclude dead cells, followed by Fc-receptor blocking and cell-surface staining with specific antibodies. The cells were analyzed using either LSR FortessaTM (BD) or Cytek[®] Aurora (CYTEK) flow cytometers. Data was obtained using either FACSDivaTM (BD) or SpectroFlo[®] (CYTEK) software and was further analyzed using FlowJoTM.

Lymphocyte stimulations

After tissue harvest, ILC stimulations were performed by incubating 0.5-1 $\times$ 10⁶ cells for 4 hours at 37°C in stimulation media (DMEM with 5% fetal bovine serum, 1% penicillin/streptomycin, 2 mM L-glutamine, 50 ng/mL Phorbol 12-myristate 13-acetate (PMA), 100 ng/mL ionomycin, 5 μ g/mL Brefeldin A (BFA), 2 μ M monensin). T cell stimulations were performed by first coating a 96-well

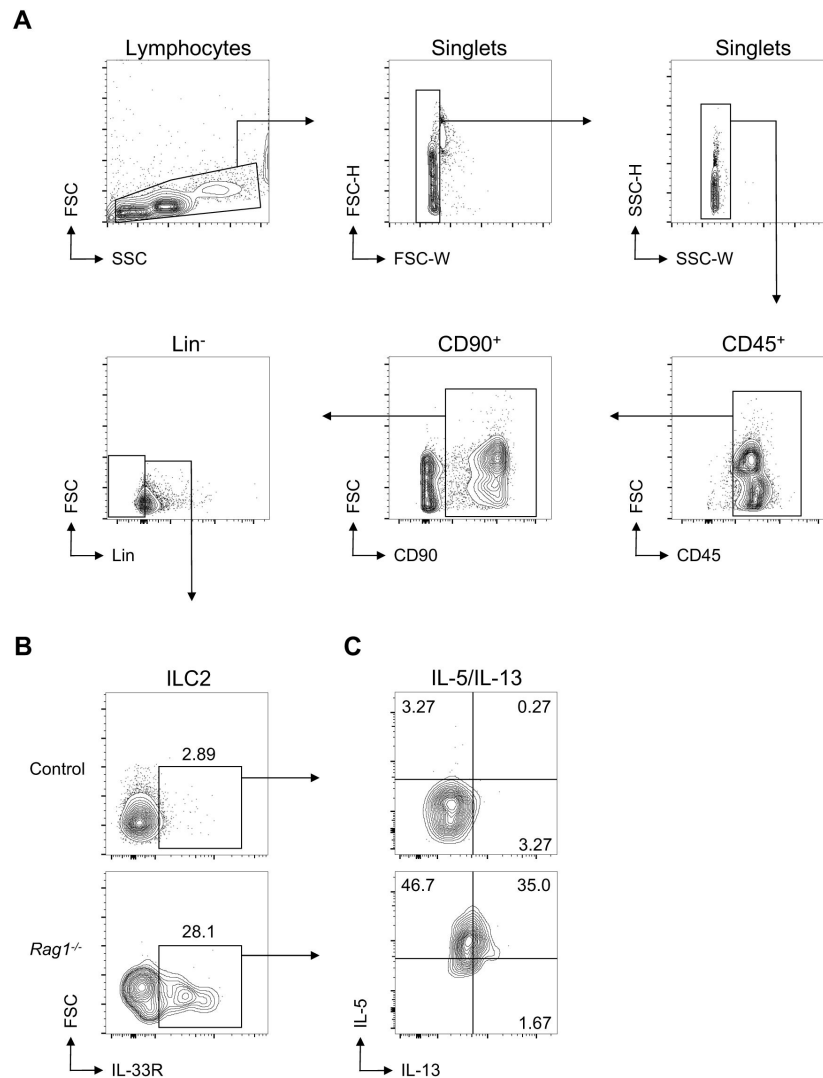


Figure S1

ILC2 and IL-5/IL-13 gating.

Gating for **A**) CD45⁺, CD90⁺, Lin⁻ cells (Lin⁻ defined as CD3⁻, CD5⁻, CD11b⁻, CD11c⁻, CD19⁻, NK1.1⁻, and FcεR1⁻), then gating on **B**) ILC2 (IL-33R⁺ Lin⁻) corresponding to **Fig. 1C**, with subsequent gating of **C**) IL-5⁺ and IL-13⁺ ILC2, corresponding to **Fig. 1D-E**.

Figure S2

Expansion and activation of ILC2s in RAG2 deficiency compared to littermates.

A) Schematic of steady state analysis of WT B6 (Control) mice or *Rag2*^{-/-} mice. **B**) Proportion of CD90⁺ Lin⁻ cells (Lin-defined as CD3⁻, CD5⁻, CD11b⁻, CD11c⁻, CD19⁻, NK1.1⁻, and FcεR1⁻) determined to be ILC2s (IL-33R⁺) in sdLN at steady state from WT or *Rag2*^{-/-} mice. Percent ILC2 from sdLN at steady state following PMA/iono stimulation positive for **C**) IL-5 or **D**) IL-13 staining. Data representative of 2 independent experiments, 2-3 mice per group. * P < 0.05, ** P < 0.01 by two-tailed Welch's t test. All data represented as mean with standard deviation.

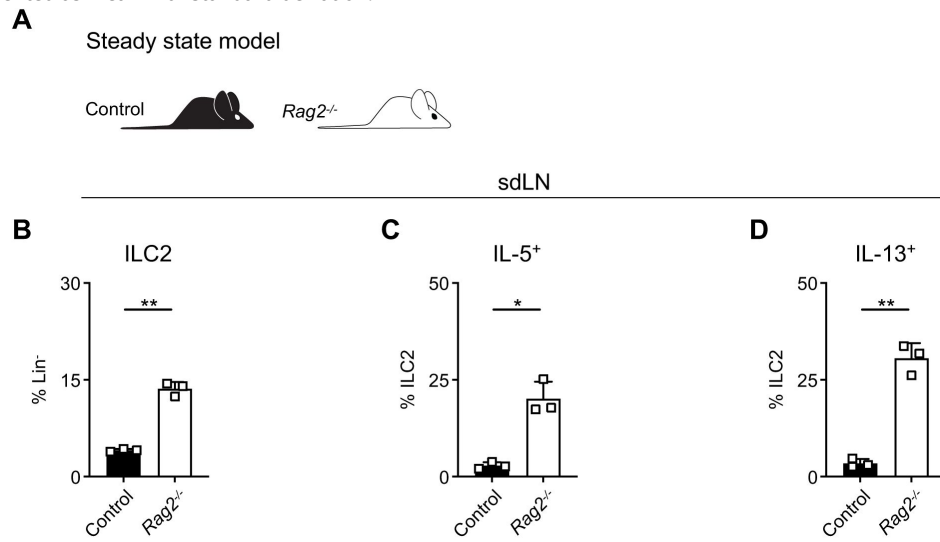
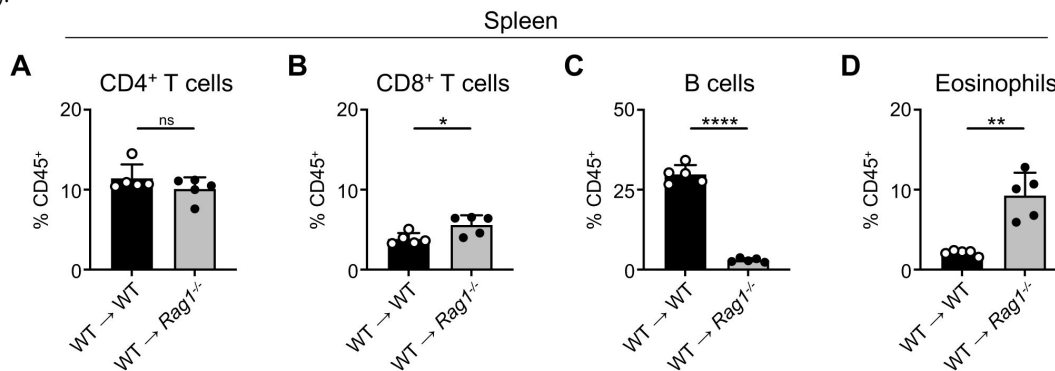


Figure S3

Confirmation of splenocyte reconstitution in splenocyte chimera mice.

Proportion of CD45⁺ splenocytes from splenocyte chimera mice (related to **Fig. 2A-E**) determined to be **A**) CD4⁺ T cells (CD4⁺, CD8⁻, CD19⁻), **B**) CD8⁺ T cells (CD4⁻, CD8⁺, CD19⁻), **C**) B cells (CD4⁻, CD8⁻, CD19⁺), and **D**) Eosinophils (SiglecF⁺, CD4⁻, CD8⁻).



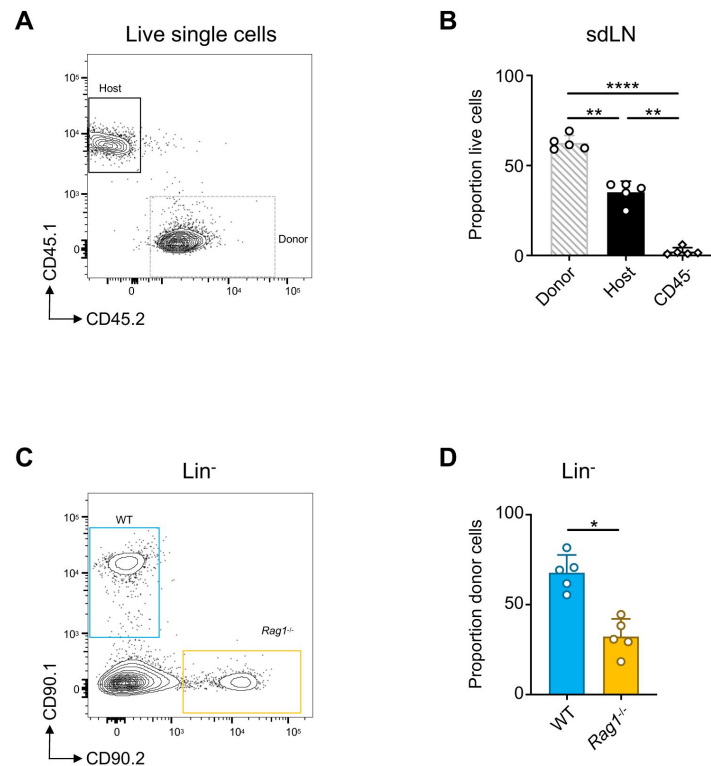


Figure S4

Donor cell reconstitution and gating in sdLN of WT:*Rag1*^{-/-} bone marrow chimera mice.

A) Gating of live host (CD45.1⁺) and donor (CD45.2⁺) cells and **B)** quantification in sdLN of WT:*Rag1*^{-/-} bone marrow chimera mice. ** $P < 0.01$, **** $P < 0.0001$, by RM one-way ANOVA test with Geisser-Greenhouse correction. **C)** Gating, and **D)** quantification of CD45.2⁺ Lin⁻ donor cells by genotype (WT – blue and *Rag1*^{-/-} – orange) in sdLN of WT:*Rag1*^{-/-} bone marrow chimera mice. * $P < 0.05$ by ratio means paired t test. All data represented as mean with standard deviation. Related to **Fig. 2F-I**.

Figure S5

sdLN multiome experiment.

A) *Rag1*^{Cre::Rosa26}^{LSL-tdRFP} reporter mice were given topical treatments with 2 nmol MC903 dissolved in ethanol vehicle or with ethanol vehicle alone to each ear daily for 7 days. Harvested sdLN processed using Magnetic Activated Cell Sorting (MACS) led to depletion of cells expressing the CD3, CD19, and CD11b lineage markers and remaining cells were further processed in the 10X Multiome pipeline, generating both single cell RNA-sequencing and single cell ATAC-sequencing data for each cell. **B)** Ear thickness measured daily in the AD-like disease multiome experiment. Data representative of one experiment, with 4 mice per group pooled for sequencing. **** P < 0.0001 by 2-way ANOVA with Sidak's multiple comparisons test, day 7. All data represented as mean with standard deviation.

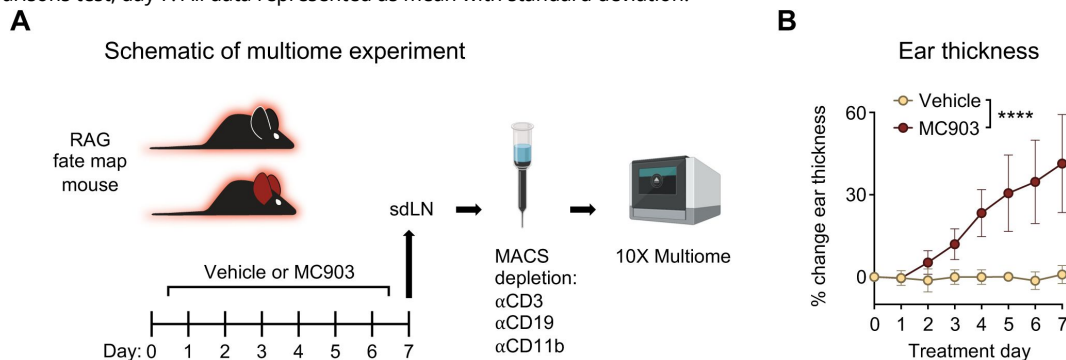
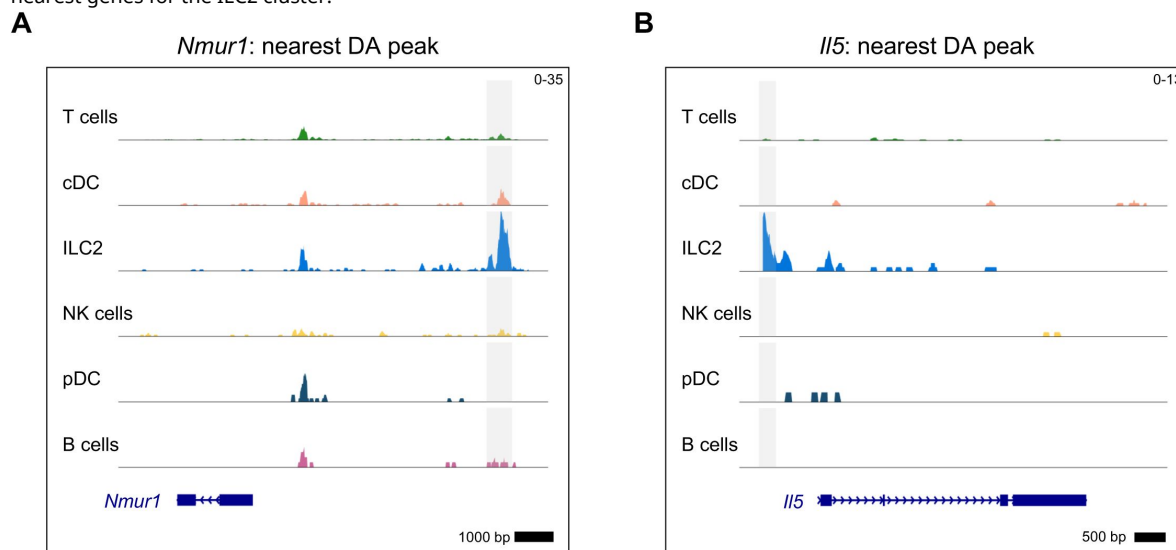


Figure S6

ILC2 marker genes identified in the differentially accessible open chromatin assay.

Differentially accessible (DA) open chromatin peaks identified for the ILC2 cluster are highlighted in gray and shown next to the closest gene for **A)** Neuromedin U receptor 1 (*Nmur1*) and **B)** IL-5 (*Il5*). See **Table S3** for top 100 DA peaks and distances to nearest genes for the ILC2 cluster.



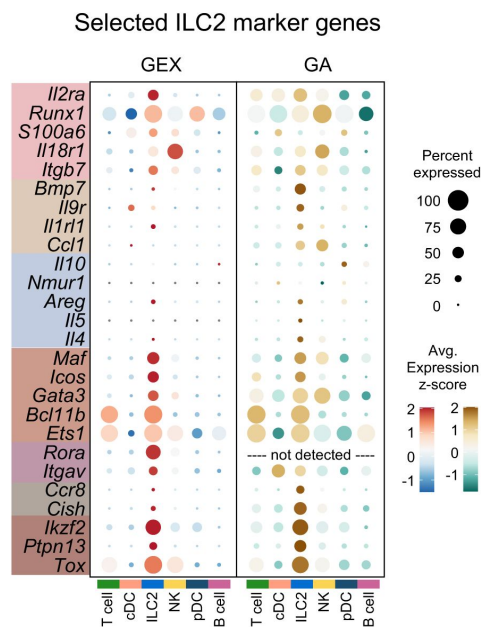


Figure S7

Dotplots of selected ILC2 marker genes.

Dotplots comparing selected marker genes from the multiomic ILC2 gene set (**Figure 4I**, **Table S4**) for each cluster between the GEX and GA assays, with genes highlighted by color corresponding to individual assays or overlap of assays in which they were identified. *Rora* was not detected in the GA assay.

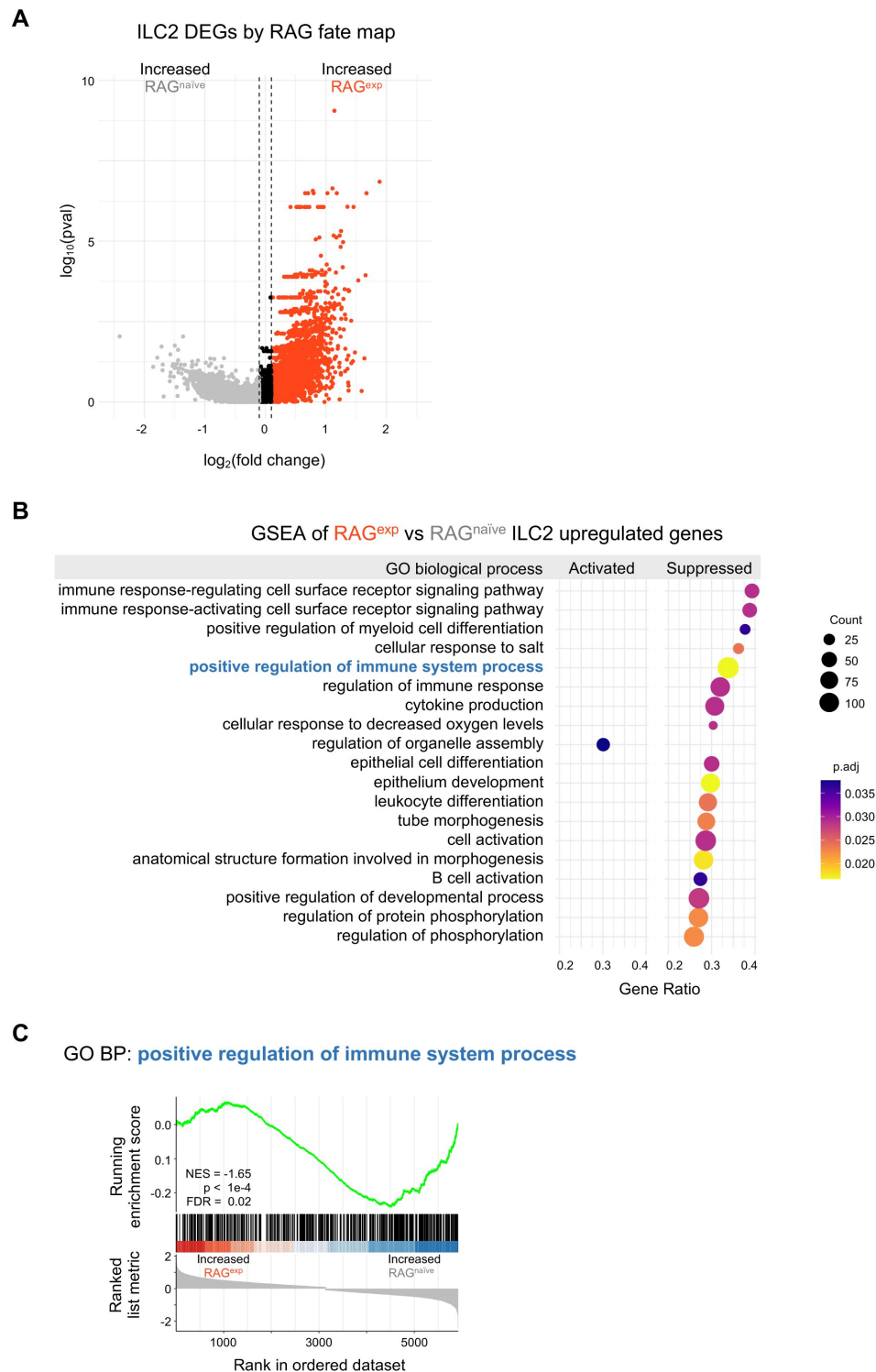


Figure S8

Gene set enrichment analysis of differentially expressed genes in ILC2s.

A) Volcano plot of differentially expressed genes (DEGs) by RAG fate map for the ILC2 cluster. A ranked list (**Table S5**) was constructed for all DEGs with $\log_2(\text{fold change}) > 0.1$ for gene set enrichment analysis (GSEA). **B)** Dotplot of GSEA result calculated using ClusterProfiler and the gene ontology (GO) biological process (BP) database (see methods). Full results in **Table S6**. **C)** GSEA plot of the GO BP “positive regulation of immune system process” gene set. **RAG^{naive}**, RAG fate map negative; **RAG^{exp}**, RAG fate map positive.

Figure S9

Mapping gene to peak links in select ILC2 genes.

Gene to peak links (GPLs) mapped for the RAG^{naïve} and RAG^{exp} states as depicted in **Figure 5B** for **A)** GATA binding protein 3 (*Gata3*) and **B)** Nedd4 family interacting protein 1 (*Ndfip1*). Only GPLs that fit in the coverage window are shown. Select peaks (teal bars) present in one state, but not the other, are highlighted in teal boxes. Full gene names not shown on figure in (A) are *9230102004Rik and **4930412013Rik and in (B) #Gm42690.

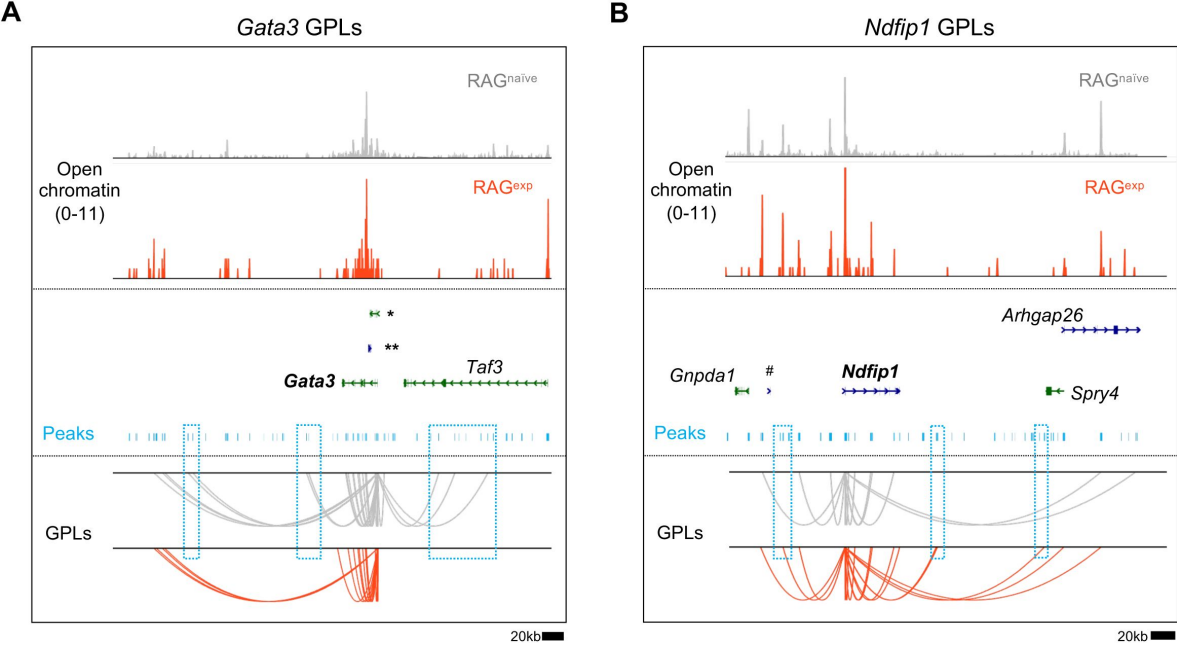


Figure S10

Gene to peak link analysis by RAG fate map and disease for all detected genes.

A) UpSet plot of overlaps in peaks identified from GPLs of all genes split by both RAG fate map (RAG^{exp} and RAG^{naïve}) and disease state (SS – steady state, AD – AD-like inflammation). Each row corresponds to one of the four sets, and each column corresponds to an intersection of one or more sets (see methods). See **Table S11** for full list of peaks from GPLs for all genes in each set. Columns identifying key intersections are color coded by the corresponding RAG fate map or disease groups. The blue column indicates the intersection of peaks from RAG^{naïve} cells and peaks induced by AD-like disease in RAG^{exp} cells. Intersection of all GPL peaks by RAG fate map and disease

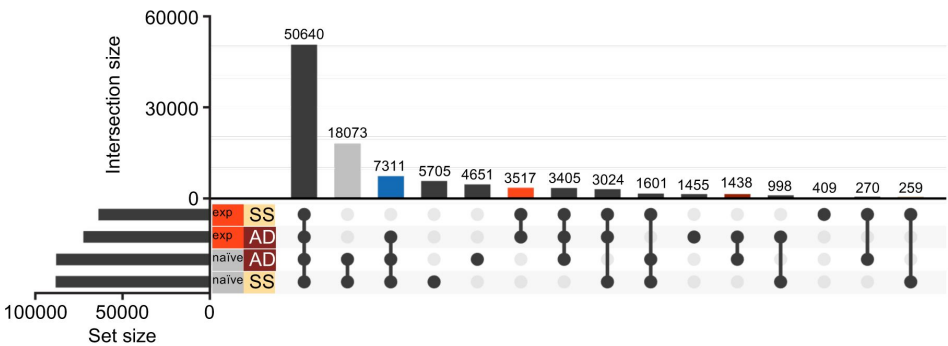


plate with 5 µg/mL anti-mouse CD3 (Biolegend) in 50 µL/well PBS overnight the day before tissue harvest. The following day, $0.5-1 \times 10^6$ cells were resuspended in 50 µL/well T cell stimulation media (5 µg/mL anti-mouse CD28 (Biolegend), 5 µg/mL BFA), 2 µM monensin) and incubated for 20 minutes hours at 37 C. The cells were then transferred to the anti-mouse CD3 coated plate and incubated for 4 hours at 37 C. After stimulation, cells were washed in wash medium, fixed, and stained for surface and intracellular markers as described for unstimulated cells.

Splenocyte chimeras

Spleens were harvested from donor WT B6 mice and immediately homogenized manually through a 100 µm cell strainer (Fisher Scientific) into a 50 mL tube with the end of a plunger from a 3 mL syringe. The strainer was washed with wash medium (2% vol/vol FBS/PBS) and the strained cells were centrifuged at 400g for 5 minutes at 4°C followed by treatment with RBC lysis buffer for two minutes and two wash steps using 2 volumes of wash medium. Cells were counted, and 5 million splenocytes were injected intraperitoneally into each recipient mouse. Experiments were performed 4 weeks following splenocyte add-back to allow immune reconstitution.

Bone marrow chimeras

Recipient mice were provided with antibiotic water, consisting of 5 mL of Sulfatrim (sulfamethoxazole/trimethoprim) added into 200 mL of drinking water, for one week starting from the day prior to irradiation (day -1). On day 0, recipient mice were irradiated with 950 cGy using the X-RAD 320 (Precision X-Ray). BM was harvested from donor mice femurs and tibias and treated with RBC lysis buffer (Sigma-Aldrich) for two minutes. BM cells were transferred into a 15 mL conical tube through a 70 µm cell strainer (Fisher Scientific) and the cell strainer and cells were washed with 2% (vol/vol) FBS/PBS. The concentration of living cells was determined using a Cellometer Auto 2000 (Nexcelom Bioscience) with ViaStain™ AOPI Staining Solution (Nexcelom Bioscience). Recipient mice received the same number of cells, at 1×10^7 live bone marrow cells per mouse, through retroorbital injection within 24-hour after irradiation. Recipients were given 8 weeks for immune reconstitution after BM transplantation before experimental use.

Cryopreserving sdLN cells for sequencing

Rag1^{Cre}::Rosa26^{LSL-tdRFP} mice were treated with 2 nmol calcipotriol (MC903, Tocris Bioscience) in 10 µL of 100% ethanol (EtOH) vehicle, or vehicle alone, on the bilateral ear skin daily for 7 days to induce AD-like inflammation. The next day, cervical sdLN were harvested and immediately homogenized manually through a 100 µm cell strainer (Fisher Scientific) into a 50 mL tube with the end of a plunger from a 3 mL syringe. The strainer was washed with wash medium (2% vol/vol FBS/PBS) and the strained cells were centrifuged at 400g for 5 minutes at 4°C. Next, cells were incubated with biotinylated antibodies (anti-mouse CD3e, CD19, CD11b; 1:300; Biolegend) in 100 µL of wash buffer for 20 minutes at 4°C, followed by two washes in 2 volumes of wash buffer. Next, no more than 10^7 cells were incubated with Streptavidin MicroBeads (Miltenyi) in 500 µL separation buffer (0.5% w/v BSA in PBS; BSA and PBS from Sigma) at 4°C for 20 minutes, then added to LD columns (Miltenyi) pre-equilibrated with separation buffer and loaded in a QuadroMACS™ Separator (Miltenyi) for negative cell selection. Remaining cells were eluted in 1 mL separation buffer and cells were centrifuged at 400g for 5 minutes at 4°C, followed by resuspension in freezing buffer (10% DMSO, Invitrogen; 20% FBS in DMEM, Sigma) and slow freezing to -80°C in a CoolCell™ LX (Corning) device.

Processing cryopreserved cells for multiome

Cryopreserved sdLN cells were processed as recommended by the 10X Genomics DemonstratedProtocol_NucleiIsolation_ATAC_GEX_Sequencing_RevC_(CG000365) instructions for primary cells without any modification to the protocol. Briefly, cells were thawed in a 37°C water bath followed by dilution into media (RPMI + 15% FBS, Sigma) and centrifugation at 400g for 5 minutes at 4°C. For each final sample (EtOH vehicle-or MC903-treated), cells were pooled from

sample from 3 individual mice. Cells were resuspended in PBS + 0.04% BSA (Sigma) and passed through a 40 μ m Flom strainer (Bel-art) followed by determination of cell concentration using the using Cellometer Auto 2000 (Nexcelom Bioscience) with ViaStain™ AOPI Staining Solution (Nexcelom Bioscience). Cells were centrifuged 5 minutes at 4°C and supernatant removed. Lysis Buffer (Tris HCl base with 0.1% Tween-20, 0.1% NP-40, 0.01% digitonin, 1 mM DTT, and 1 U/ μ L Protectors RNase inhibitor, Sigma; full recipe in 10X protocol) was added, cells mixed 10x, and incubated on ice for 3 minutes. Nuclei from lysed cells were centrifuged at 400g for 5 minutes at 4°C and washed in 1 mL Wash Buffer (Lysis Buffer, but without NP-40 or digitonin). The wash step was repeated two more times. Nuclei concentration was determined as for cell concentration using the Cellometer and ViaStain™ solution. The AOPI staining indicated 97-99% lysis efficiency of the cells. We manually confirmed nuclei count using a Bright-Line™ hemacytometer (Hausser Scientific™). Nuclei were centrifuged at 400g for 5 minutes and resuspended in a volume of 1X Nuclei Buffer (10X Genomics) to yield roughly 4,000 nuclei/ μ L. We then immediately proceeded to the 10X Chromium Next GEM Single Cell Multiome ATAC + Gene Expression pipeline.

Multiome library construction and sequencing

Multiome 3v3.1 GEX and ATAC libraries were prepared as recommended by 10X Genomics protocol Chromium_NextGEM_Multiome_ATAC_GEX_User_Guide_RevD ((CG000338). For sample preparation on the 10x Genomics platform, the Chromium Next GEM Single Cell Multiome ATAC + Gene Expression Reagent Bundle, 16 rxns PN-1000283, Chromium Next GEM Chip J Single Cell Kit, 48 rxns PN-1000234, Single Index Kit N Set A, 96 rxns PN-1000212 (ATAC), Dual Index Kit TT Set A, 96 rxns PN-1000215 (3v3.1 GEX), were used. The concentration of each library was accurately determined through qPCR utilizing the KAPA library Quantification Kit according to the manufacturer's protocol (KAPA Biosystems/Roche) to produce cluster counts appropriate for the Illumina NovaSeq6000 instrument. GEX libraries were pooled and run over 0.05 of a NovaSeq6000 S4 flow cell using the XP workflow and running a 28×10×10×150 sequencing recipe in accordance with manufacturer's protocol. Target coverage was 500M reads per sample. ATAC libraries were pooled and run over 0.167 of a NovaSeq6000 S1 flow cell using the XP workflow and running a 51×8×16×51 sequencing recipe in accordance with manufacturer's protocol. Target coverage was 250M reads per sample.

Multiomic data analysis

The cellranger-arc-2.0.0 (10X Genomics) pipeline was used to generate FASTQ files, gene expression matrices, and ATAC fragment tables for each sample, followed by aggregation using the aggr function. Default settings were utilized, with the exception that we incorporated a custom reference with the sequence for tdRFP (see **Supplemental file S1**) added to the default mouse reference sequence provided by cellranger (refdata-cellranger-arc-mm10-2020-A-2.0.0). Correction for ambient RNA was performed using SoupX¹²³ with clustering information provided by the default cellranger outputs. Doublets were removed using Scrublet¹²⁴ with default settings.

Corrected data was then processed using Signac³⁷ and Seurat^{34–36}. ATAC-seq peaks were identified using MACS2¹²⁵ through the CallPeaks function in Signac. Per-cell quality control metrics were computed using the TSSEnrichment and NucleosomeSignal functions, and cells retained with a nucleosome signal score < 1.5, TSS enrichment score > 1, total RNA counts < 15,000 and > 1,000, total ATAC counts < 75,000 and > 100, percent mitochondrial reads < 5%, and percent ribosomal genes detected <10%. After these filtering steps, 10,304 cells remained. Cells were further filtered by their expression of lineage defining markers similar to the negative selection step during sample processing. Cells with detectable transcripts for *Cd3d*, *Cd3e*, *Cd3g*, *Cd4*, *Cd19*, *Cd8a*, and *Itgam* were removed. This left 2,034 remaining cells for further analysis.

The SCTransform function of Seurat was used to normalize RNA counts. We performed integration of the two samples using the RNA assay to correct for batch effects and treatment in the initial clustering using the default parameters for the functions SelectIntegrationFeatures,

FindIntegrationAnchors, and IntegrateData. The integrated data was used for PCA (25 dimensions) and UMAP reduction for the RNA assay alone. With default parameters in Signac, we used TFIDF to normalize ATAC peaks and latent semantic indexing (LSI) to reduce the dimensionality of the ATAC data. We constructed a UMAP of the ATAC data alone using the LSI reduction (dimensions 2-25). To construct a joint graph and UMAP using equal weighting from the RNA and ATAC assays, we used the FindMultiModalNeighbors function of Seurat/Signac using default parameters (RNA dimensions 1-25, ATAC dimensions 2-25). We used a resolution of 0.1 to identify clusters with the FindClusters function in Seurat/Signac. Cell types were assigned based on manual curation of marker genes. Initially, 7 clusters were identified, but two highly similar lymphocyte clusters were merged for a total of 6 cell types.

The inferred Gene Activity (GA) assay from the ATAC-seq data was calculated using default parameters of the GeneActivity function in Signac. FindAllMarkers was used to identify top markers by cluster for both RNA gene expression data (GEX) and GA, with setting adjustments including min.pct = 0.20 and logfc.threshold = 0.25. The differentially accessible (DA) open chromatin assay was calculated in Signac with the FindMarkers function on the ATAC-seq peaks assay (called using MACS2 as above). The differential test used was 'LR' (logistical regression, as suggested for snRNA-seq¹²⁶). The total number of ATAC fragments was used as a latent variable to mitigate effect of differential sequencing depth. Given the sparsity of the data, the min.pct parameter was set to 0.02. After identifying the top differentially accessible peaks for each cluster, the gene closest to each peak was determined using the ClosestFeature function in Signac. Results were filtered for genes within 10⁵ base pairs of the corresponding peak. The filtered gene lists were used for the "DA" assay as markers of each cluster (top 25) and an expanded list for the ILC2 cluster (top 100). Venn diagrams were calculated using BioVenn/BioVennR¹²⁷.

Gene set enrichment analysis was performed and visualized using ClusterProfiler¹²⁸. For GSEA on steady state ILC2 DEGs between fate mapped states, we opted to use more permissive filtering parameters instead of default parameters. We created the ranked list of DEGs using the FindMarkers function in Seurat with min.pct = 0.1 and logfc.threshold = 0.1. The DEG list from the GEX assay was used to generate the GSEA results. The DEG list from the GA assay did not yield any significant GSEA results. The ClusterProfiler function gseGO was used to analyze the ranked DEG list using the parameters minGSSize = 50, maxGSSize = 500, pvalueCutoff = 0.05.

The correlation coefficients, or gene to peak links (GPLs), between gene expression and accessibility of each peak were calculated for all peaks within 10⁶ base pairs of the transcription start sites for all detected genes using the LinkPeaks function of Signac with min.cells = 2. GPLs were filtered by gene for the curated ILC2 and Th2 gene sets. Since multiple genes can be linked to one peak by GPL analysis, finding intersections of GPLs in set analysis would result in counting some epigenomic regions multiple times. Thus, for set analysis, we eliminated GPLs with redundant peaks. Then, we used each list of non-redundant peaks as input sets to generate UpSet plots and lists of intersecting peaks between states (Rag1 fate map positive or negative; AD-like disease or steady state) using UpSetR¹²⁹. Coverage plots of the single cell multiomic data, including open chromatin, peaks, and links (GPLs), were plotted using the CoveragePlot function in Signac.

Data and code availability

Sequencing data have been deposited at GEO and accession numbers are listed in the key resources table. Aggregated data are supplied in the supplemental file. All data reported in this paper will be shared by the lead contact upon request. This paper does not report original code. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
anti-mouse CD11b	eBioscience	Cat#: 45-0112-82; RRID: AB_953558
anti-mouse CD11c	eBioscience	Cat#: 45-0114-82; RRID: AB_925727
anti-mouse CD19	eBioscience	Cat#: 45-0193-82; RRID: AB_1106999
anti-mouse CD3e	eBioscience	Cat#: 45-0031-82; RRID: AB_1107000
anti-mouse CD5	eBioscience	Cat#: 45-0051-82; RRID: AB_914334
anti-mouse NK1.1	eBioscience	Cat#: 45-5941-82; RRID: AB_914361
anti-mouse FcεR1α	eBioscience	Cat#: 11-5898-82; RRID: AB_465308
anti-mouse IL-13	eBioscience	Cat#: 48-7136-42; RRID: AB_2784729
anti-mouse CD45.1	BioLegend	Cat#: 110706; RRID: AB_313495
anti-mouse CD45.2	BioLegend	Cat#: 109808; RRID: AB_313445
anti-mouse ICOS	BioLegend	Cat#: 313506; RRID: AB_416330
anti-mouse CD62L	BioLegend	Cat#: 104428; RRID: AB_830799
anti-mouse CD69	BioLegend	Cat#: 104530; RRID: AB_2563062
anti-mouse KLRG1	BioLegend	Cat#: 138424; RRID: AB_2564051
anti-mouse IL-5	BioLegend	Cat#: 504306; RRID: AB_315330
anti-mouse IL-13	BioLegend	Cat#: 503826; RRID: AB_2650897
anti-mouse IL-17A	BioLegend	Cat#: 506927; RRID: AB_11126144
anti-mouse IFNγ	BioLegend	Cat#: 505806; RRID: AB_315400
anti-mouse CD90.2	BioLegend	Cat#: 105328; RRID: AB_10613293
anti-mouse IL-33Rα	BioLegend	Cat#: 145308; RRID: AB_2565569
anti-mouse IL-33Rα	BioLegend	Cat#: 145327; RRID: AB_2565569
anti-mouse CD8a	BioLegend	Cat#: 100762; RRID: AB_2564027
anti-mouse CD25	BioLegend	Cat#: 102016; RRID: AB_312865
anti-mouse CD45.2	BioLegend	Cat#: 109806; RRID: AB_313443
anti-mouse CD90.2	BioLegend	Cat#: 109830; RRID: AB_1186098
anti-mouse CD4	BioLegend	Cat#: 100449; RRID: AB_2564587
anti-mouse Gata3	BioLegend	Cat#: 653814; RRID: AB_2563221
anti-mouse CD25	BioLegend	Cat#: 102036; RRID: AB_2563059
anti-mouse CD90.1	BioLegend	Cat#: 202537; RRID: AB_2562644
anti-mouse CD3e	BioLegend	Cat#: 155608; RRID: AB_2750434
anti-mouse CD3e	BioLegend	Cat#: 100339; RRID: AB_11150783
anti-mouse CD28	BioLegend	Cat#: 102115; RRID: AB_11150408
anti-mouse CD3	BioLegend	Cat#: 100243; RRID: AB_2563946
anti-mouse CD19	BioLegend	Cat#: 115503; RRID: AB_313638
anti-mouse CD11b	BioLegend	Cat#: 101203; RRID: AB_312786
anti-mouse IL-4	BioLegend	Cat#: 504109; RRID: AB_493320
anti-mouse CD117 (c-Kit)	BioLegend	Cat#: 105838; RRID: AB_2616739
anti-mouse CD49b	Invitrogen	Cat#: 17-5971-82; RRID: AB_469485
anti-mouse CD45R/B220	BioLegend	Cat#: 103275; RRID: AB_2860602
anti-mouse I-A/I-E (MHCII)	BioLegend	Cat#: 107622; RRID: AB_493727
anti-mouse Ly-6A/E (Sca-1)	BioLegend	Cat#: 122512; RRID: AB_756197
anti-mouse F4/80	BioLegend	Cat#: 123112; RRID: AB_893482

anti-mouse SiglecF	BD Biosciences	Cat#: 562757; RRID: AB_2687994
anti-mouse TCR γ/δ	eBioscience	Cat#: 48-5711-82; RRID: AB_2574071
anti-mouse Ly-6G	eBioscience	Cat#: 62-9668-82; RRID: AB_2762763
anti-mouse CD16/CD32	Bio X Cell	Cat#: BE0307; RRID: AB_1107647
streptavidin	BioLegend	Cat#: 405204
streptavidin	BioLegend	Cat#: 405207
Chemicals, peptides, and recombinant proteins		
Phorbol 12-myristate 13-acetate (PMA)	Sigma	P1585
Ionomycin	Sigma	I0634
Monensin	Biolegend	420701
Brefeldin A Solution	Biolegend	420601
DMEM	Sigma	D6429
RPMI-1640 Medium	Sigma	R8758
Bovine serum albumin (BSA)	Sigma	A2058
Penicillin-Streptomycin	Gibco	15140122
L-glutamine solution - 200 mM	Corning	MT25005CI
Fetal bovine serum	Sigma	2442
Calcipotriol (MC903)	Tocris Biosciences	2700
Phosphate Buffered Saline	Sigma	806552
DMSO	Invitrogen	D12345
Nuclei Buffer (20X)	10x Genomics	2000153/2000207
Digitonin	ThermoFisher	BN2006
Trizma hydrochloride solution pH 7.4	Sigma	T2194
Sodium Chloride Solution 5 M	Sigma	59222C
Magnesium Chloride Solution 1M	Sigma	M1028
Nonidet P40 Substitute	Sigma	74385
Protector RNase inhibitor	Sigma	3335402001
DTT	Sigma	646563
Flowmi Cell Strainer 40 μ m	Bel-Art	H13680-0040
Tween 20	Bio-Rad	1662404
ZombieNIR	Biolegend	423106
ZombieUV	Biolegend	423107
Experimental models: Organisms/strains		
B6 WT	Jackson Laboratory	C57BL/6J; Cat# 000664
Rag1 ^{-/-}	Jackson Laboratory	B6.129S7-Rag1tm1Mom/J; Cat# 002216
Rag2 ^{-/-}	Jackson Laboratory	B6.Cg-Rag2tm1.1Cgn/J; Cat# 002014
B6 CD45.1	Jackson Laboratory	B6.SJL-Ptprca Pepcb/BoyJ; Cat# 008450
B6 CD90.1	Jackson Laboratory	B6.PL-Thy1a/CyJ; Cat# 000406
Rag1 fate-map mice	^{24,33}	

Software and algorithms		
BD FACSDiva (v8.0)	BD Life Sciences	https://www.bdbiosciences.com/en-us/products/software/instrument-software/bd-facsdiva-software
FlowJo™ (v10.8)	BD Life Sciences	https://www.flowjo.com/
SpectroFlo® (v3.0)	CYTEK	https://cytekbio.com/pages/spectroflo
Prism 9	GraphPad Software	https://www.graphpad.com/scientific-software/prism/
R (v4.2.2)	R core	https://www.r-project.org/
Seurat (v4.2.0)	35	https://github.com/satijalab/seurat
Signac (v1.8.0)	37	https://github.com/stuart-lab/signac
chromVAR (v1.18.0)	130	https://github.com/GreenleafLab/chromVAR
JASPAR2022 (v0.99.7)		https://github.com/dabar/JASPAR2022
SoupX (v1.6.1)	123	https://github.com/constantAmateur/SoupX
clusterProfiler (v4.4.4)	128	https://github.com/YuLab-SMU/clusterProfiler
singleCellITK (v2.6.0)	131	https://github.com/compbiomed/singleCellITK
biomaRt (v2.52.0)	132	https://github.com/grimbough/biomaRt
EnsDb.Mmusculus.v79 (v2.99.0)	133	https://bioconductor.org/packages/release/data/annotation/html/EnsDb.Mmusculus.v79.html
BSgenome.Mmusculus.UCSC.mm10 (v1.4.3)	134	https://bioconductor.org/packages/release/data/annotation/html/BSgenome.Mmusculus.UCSC.mm10.html
ggplot2 (v3.3.6)	135	https://github.com/tidyverse/ggplot2
viridis (v0.6.2)	136	https://github.com/sjmgarnier/viridis
TFBSTools (v1.34.0)	137	https://bioconductor.org/packages/release/bioc/html/TFBSTools.html
motifmatchr (v1.18.0)	138	https://github.com/GreenleafLab/motifmatchr
BioVenn/BioVennR	127	https://www.biovenn.nl/
UpSetR	129	http://gehlenborglab.org/research/projects/upsetr/
Cell Ranger	139,140	https://support.10xgenomics.com/single-cell-multiome-atac-gex/software/overview/welcome
Python (v3.7)	Python Software Foundation	https://www.python.org/
Deposited data		
scRNA-seq dataset	This paper	GEO: GSE192597

scATAC-seq dataset	This paper	GEO: GSE192597
Other		
Streptavidin MicroBeads	Miltenyi	130-048-101
LD Columns	Miltenyi	130-042-901
QuadroMACS™ Separator	Miltenyi	130-090-976
CoolCell™ LX	Corning	CLS432001
Bright-Line™ hemacytometer	Hausser Scientific	3110V
Cytek® Aurora	CYTEK	N/A
LSR Fortessa™	BD Life Sciences	N/A

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Editors

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Reviewer #1 (Public Review):

The study starts with the notion that in an AD-like disease model, ILC2s in the Rag1 knock-out were expanded and contained relatively more IL-5+ and IL-13+ ILC2s. This was confirmed in the Rag2 knock-out mouse model.

By using a chimeric mouse model in which wild-type knock-out splenocytes were injected into irradiated Rag1 knock-out mice, it was shown that even though the adaptive lymphocyte compartment was restored, there were increased AD-like symptoms and increased ILC2 expansion and activity. Moreover, in the reverse chimeric model, i.e. injecting a mix of wild-type and Rag1 knock-out splenocytes into irradiated wild-type animals, it was shown that the Rag1 knock-out ILC2s expanded more and were more active. Therefore, the authors could conclude that the RAG1 mediated effects were ILC2 cell-intrinsic.

Subsequent fate-mapping experiments using the Rag1Cre;reporter mouse model showed that there were indeed RAGnaïve and RAGexp ILC2 populations within naïve mice. Lastly, the authors performed multi-omic profiling, using single-cell RNA sequencing and ATAC-sequencing, in which a specific gene expression profile was associated with ILC2. These included well-known genes but the authors notably also found expression of *Ccl1* and *Ccr8* within the ILC2. The authors confirmed their earlier observations that in the RAGexp ILC2 population, the Th2 regulome was more suppressed, i.e. more closed, compared to the RAGnaïve population, indicative of the suppressive function of RAG on ILC2 activity. I do agree with the authors' notion that the main weakness was that this study lacks the mechanism by which RAG regulates these changes in ILC2s.

The manuscript is very well written and easy to follow, and the compelling conclusions are well supported by the data. The experiments are meticulously designed and presented. I wish to commend the authors for the study's quality.

Even though the study is compelling and well supported by the presented data, some additional context could increase the significance:

(1) The presence of the RAGnaïve and RAGexp ILC2 populations raises some questions on the (different?) origin of these populations. It is known that there are different waves of ILC2 origin (most notably shown in the Schneider et al Immunity 2019 publication, PMID 31128962). I believe it would be very interesting to further discuss or possibly show if there are different origins for these two ILC populations.

Several publications describe the presence and origin of ILC2s in/from the thymus (PMIDs 33432227 24155745). Could the authors discuss whether there might be a common origin for the RAGexp ILC2 and Th2 cells from a thymic lineage? If true that the two populations would be derived from different populations, e.g. being the embryonic (possibly RAGnaïve) vs. adult bone marrow/thymus (possibly RAGexp), this would show a unique functional difference between the embryonic derived ILC2 vs. adult ILC2.

(2) On line 104 & Figures 1C/G etc. the authors describe that in the RAG knock-out ILC2 are relatively more abundant in the lineage negative fraction. On line 108 they further briefly mentioned that this observation is an indication of enhanced ILC2 expansion. Since the study includes an extensive multi-omics analysis, could the authors discuss whether they have seen a correlation of RAG expression in ILC2 with regulation of genes associated with proliferation, which could explain this phenomenon?

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Reviewer #2 (Public Review):**Summary:**

The study by Ver Heul et al., investigates the consequences of RAG expression for type 2 innate lymphoid cell (ILC2) function. RAG expression is essential for the generation of the receptors expressed by B and T cells and their subsequent development. Innate lymphocytes, which arise from the same initial progenitor populations, are in part defined by their ability to develop in the absence of RAG expression. However, it has been described in multiple studies that a significant proportion of innate lymphocytes show a history of Rag expression. In compelling studies several years ago, members of this research team revealed that early Rag expression during the development of Natural Killer cells (Karo et al., Cell 2014), the first described innate lymphocyte, had functional consequences.

Here, the authors revisit this topic, a worthwhile endeavour given the broad history of Rag expression within all ILCs and the common use of RAG-deficient mice to specifically assess ILC function. Focusing on ILC2s and utilising state-of-the-art approaches, the authors sought to understand whether early expression of Rag during ILC2 development had consequences for activity, fitness, or function. Having identified cell-intrinsic effects in vivo, the authors investigated the causes of this, identifying epigenetic changes associated with the accessibility genes associated with core ILC2 functions.

The manuscript is well written and does an excellent job of supporting the reader through reasonably complex transcriptional and epigenetic analyses, with considerate use of explanatory diagrams. Overall I think that the conclusions are fair, the topic is thought-provoking, and the research is likely of broad immunological interest. I think that the extent of functional data and mechanistic insight is appropriate.

Strengths:

- The logical and stepwise use of mouse models to first demonstrate the impact on ILC2 function in vivo and a cell-intrinsic role. Initial analyses show enhanced cytokine production by ILC2 from RAG-deficient mice. Then through two different chimeric mice (including BM chimeras), the authors convincingly show this is cell intrinsic and not simply as a result of lymphopenia. This is important given other studies implicating enhanced ILC function in RAG-/- mice reflect altered competition for resources (e.g. cytokines).
- Use of Rag expression fate mapping to support analyses of how cells were impacted - this enables a robust platform supporting subsequent analyses of the consequences of Rag expression for ILC2.
- Use of snRNA-seq supports gene expression and chromatin accessibility studies - these reveal clear differences in the data sets consistent with altered ILC2 function.
- Convincing evidence of epigenetic changes associated with loci strongly linked to ILC2 function. This forms a detailed analysis that potentially helps explain some of the altered ILC2 functions observed in ex vivo stimulation assays.
- Provision of a wealth of expression data and bioinformatics analyses that can serve as valuable resources to the field.

Weaknesses:

- Lack of insight into precisely how early RAG expression mediates its effects, although I think this is beyond the scale of this current manuscript. Really this is the fundamental next question from the data provided here.

- The epigenetic analyses provide evidence of differences in the state of chromatin, but there is no data on what may be interacting or binding at these sites, impeding understanding of what this means mechanistically.
- Focus on ILC2 from skin-draining lymph nodes rather than the principal site of ILC2 activity itself (the skin). This may well reflect the ease at which cells can be isolated from different tissues.
- Comparison with ILC2 from other sites would have helped to substantiate findings and compensate for the reliance on data on ILC2 from skin-draining lymph nodes, which are not usually assessed amongst ILC2 populations.
- The studies of how ILC2 are impacted are a little limited, focused exclusively on IL-13 and IL-5 cytokine expression.

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Reviewer #3 (Public Review):

In this study, Ver Heul et al. investigate the role of RAG expression in ILC2 functions. While RAG genes are not required for the development of ILCs, previous studies have reported a history of expression in these cells. The authors aim to determine the potential consequences of this expression in mature cells. They demonstrate that ILC2s from RAG1 or RAG2 deficient mice exhibit increased expression of IL-5 and IL-13 and suggest that these cells are expanded in the absence of RAG expression. However, it is unclear whether this effect is due to a direct impact of RAG genes or a consequence of the lack of T and B cells in this condition. This ambiguity represents a key issue with this study: distinguishing the direct effects of RAG genes from the indirect consequences of a lymphopenic environment.

The authors focus their study on ILC2s found in the skin-draining lymph nodes, omitting analysis of tissues where ILC2s are more enriched, such as the gut, lungs, and fat tissue. This approach is surprising given the goal of evaluating the role of RAG genes in ILC2s across different tissues. The study shows that ILC2s derived from RAG^{-/-} mice are more activated than those from WT mice, and RAG-deficient mice show increased inflammation in an atopic dermatitis (AD)-like disease model. The authors use an elegant model to distinguish ILC2s with a history of RAG expression from those that never expressed RAG genes. However, this model is currently limited to transcriptional and epigenomic analyses, which suggest that RAG genes suppress the type 2 regulome at the Th2 locus in ILC2s.

The authors report a higher frequency of ILC2s in RAG^{-/-} mice in skin-draining lymph nodes, which is expected as these mice lack T and B cells, leading to ILC expansion. Previous studies have reported hyper-activation of ILCs in RAG-deficient mice, suggesting that this is not necessarily an intrinsic phenomenon. For example, RAG^{-/-} mice exhibit hyperphosphorylation of STAT3 in the gut, leading to hyperactivation of ILC3s. This study does not currently provide conclusive evidence of an intrinsic role of RAG genes in the hyperactivation of ILC2s. The splenocyte chimera model is artificial and does not reflect a normal environment in tissues other than the spleen. Similarly, the mixed BM model does not demonstrate an intrinsic role of RAG genes, as RAG1^{-/-} BM cells cannot contribute to the B and T cell pool, leading to an expected expansion of ILC2s. As the data are currently presented it is expected that a proportion of IL-5-producing cells will come from the RAG1^{-/-} BM.

Overall, the level of analysis could be improved. Total cell numbers are not presented, the response of other immune cells to IL-5 and IL-13 (except the eosinophils in the splenocyte chimera mice) is not analyzed, and the analysis is limited to skin-draining lymph nodes.

The authors have a promising model in which they can track ILC2s that have expressed RAG or not. They need to perform a comprehensive characterization of ILC2s in these mice, which develop in a normal environment with T and B cells. Approximately 50% of the ILC2s have a history of RAG expression. It would be valuable to know whether these cells differ from ILC2s that never expressed RAG, in terms of proliferation and expression of IL-5 and IL-13. These analyses should be conducted in different tissues, as ILC2s adapt their phenotype and transcriptional landscape to their environment. Additionally, the authors should perform their AD-like disease model in these mice.

The authors provide a valuable dataset of single-nuclei RNA sequencing (snRNA-seq) and ATAC sequencing (snATAC-seq) from RAGexp (RAG fate map-positive) and RAGnaïve (RAG fate map-negative) ILC2s. This elegant approach demonstrates that ILC2s with a history of RAG expression are epigenomically suppressed. However, key genes such as IL-5 and IL-13 do not appear to be differentially regulated between RAGexp and RAGnaïve ILC2s according to Table S5. Although the authors show that the regulome activity of IL-5 and IL-13 is decreased in RAGexp ILC2s, how do the authors explain that these genes are not differentially expressed between the RAGexp and RAGnaïve ILC2? I think that it is important to validate this in vivo.

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